

170 Ventricular Tumors

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This chapter includes an accompanying lecture presentation that has been prepared by the authors: Video 170.1.

KEY CONCEPTS

- Ventricular tumors are rare, making up 1.6% of intracranial tumors. Tumors are classified as either primary, arising from structures of or within the ventricle, or secondary, originating from paraventricular regions.
- Presenting symptoms of ventricular tumors are typically insidious and are caused by local tumor invasion into functional tissue, increased intracranial pressure (ICP) from the tumor mass, or hydrocephalus (obstructive more often than communicating).
- Preoperative diagnostic studies are key to developing a treatment plan as some tumors are best treated noninvasively. If surgery is chosen, vascular imaging is particularly important for choosing the proper approach.
- Intimate knowledge of surface and deep anatomy is necessary to utilize or generate safe operative corridors that avoid eloquent cortical regions and functional networks.
- Dynamic retraction significantly reduces the risk for cortical ischemia or venous infarction. Effective preoperative planning takes advantage of gravity and early cisternal or ventricular access to allow for cerebral relaxation.
- Surgeons should be comfortable with all techniques and approaches described in this chapter to select the best approach for a patient's unique anatomy and pathology.
- Neuroendoscopy is a rapidly growing technology that provides remarkable visualization of deep-seated lesions leveraging the use of high-definition cameras, angled lenses, a powerful light source, and the ability to bypass superficial visual obstructions.
- Novel targeted therapies, such as BRAF and MEK inhibitors for treatment of papillary craniopharyngioma, are poised to significantly alter management plans, however, long-term data regarding durability and side effects continue to emerge.

In the third century BC, the Greek physician-anatomist Herophilus established the oldest known documentation of the cerebral ventricles in his description of the “four small stomachs” of the brain and their communications.¹ In 1504, Leonardo da Vinci performed the first cerebral ventriculography by injecting molten wax into the ventricles of an ox, resulting in a cast that he used to draw their anatomy and conjecture as to their function.² However, it was not until 1764 that Italian physician Domenico Felice Antonio Cotugno defined CSF while conducting a novel method of autopsy performed with the patient upside down.

Any account of the history of ventricular surgery would be amiss to not mention the contributions of Walter Dandy, who is often credited as the father of ventricular surgery and neuroendoscopy. A vocal and early proponent of intraventricular surgery, his contributions in this endeavor include pneumoencephalography, third ventriculostomy, choroid plexus extirpation, and cannulation of the sylvian aqueduct. He opened the field to intraventricular surgery with a series of early successes, including removal of the first benign lateral ventricular mass, followed soon after by resection of a third ventricular colloid cyst. Throughout his career, Dandy authored more than 30 publications describing intraventricular tumors and their removal. In 1933, he compiled his experiences into a single book, *Benign Tumors in the Third Ventricle of the Brain: Diagnosis and Treatment*.³

Today, intraventricular surgery is safer and less invasive as a result of modern technologies, advanced imaging, improved microsurgical technique, increased operator comfort with complex approaches, and the development of neuroanesthesia. As contemporary neurosurgeons have become increasingly facile with endoscopic surgical techniques, Dandy's vision of surgeries performed entirely with an endoscope has become reality. It is because of Dandy's pioneering work and innovative spirit that ventricular tumor surgery—once a surgical no man's land—exists in the modern neurosurgeon's armamentarium.

GENERAL EPIDEMIOLOGY AND CLASSIFICATION

Ventricular tumors encompass a wide assortment of pathologies occupying a common anatomic space: the cerebral ventricles (Table 170.1). Tumors in this location are rare, accounting for less than 1.6% of all intracranial lesions.^{4–8} Pecker initially described how these tumors could be separated into two categories based on their origin: primary or secondary.⁹ Primary (“true intraventricular”) tumors originate within the ventricles from either the ependymal lining of the ventricular wall, choroid plexus, or tela choroidea. Secondary (“paraventricular”) tumors originate outside the ventricles but have an exophytic component extending into the ventricle. Metastatic lesions may arise in either location. Understanding the origin of these tumors has important diagnostic and surgical implications, which will be discussed later.

Clinical Presentation

Often benign and slow-growing, ventricular tumors may reach a substantial size before detection.¹⁰ Lesions become symptomatic as a result of (1) invasion of local structures, (2) increased intracranial pressure (ICP) from the tumor mass, or (3) hydrocephalus. Presenting symptoms in adults include headache, lethargy, nausea, memory loss, gait disorder, and cognitive or mood changes.^{4,11,12} Children may present with similar findings in addition to delayed developmental milestones or retarded growth. Increased head circumference, a bulging fontanelle, distended scalp veins, and failure to thrive classically manifest in the infant. Despite their often insidious presentation, acute deterioration can occur by means of sudden CSF pathway obstruction or intralesional hemorrhage, resulting in a rapid and severe onset of symptoms.

Communicating hydrocephalus may be seen with specific pathologies caused by overproduction of CSF (e.g., choroid plexus

TABLE 170.1 Common Ventricular Tumors

Ventricular Pathology	WHO Grade	Most Common Location(s)
NEURONAL AND NEUROGLIAL TUMORS		
Diffuse astrocytic and oligodendroglial tumors	I–IV	Lateral and third ventricles
Pilocytic astrocytomas	I	Fourth ventricle, suprasellar
Subependymal giant cell astrocytoma (SEGA)	I	Lateral ventricles
Central neurocytoma	II	Lateral ventricles
Chordoid glioma of the third ventricle	II	Third ventricle
EPENDYMAL TUMORS		
Subependymoma	I	Fourth and lateral ventricles
Ependymoma	II or III	Fourth and lateral ventricles
CHOROID PLEXUS TUMORS		
Choroid plexus papilloma	I	Fourth and lateral ventricles
Atypical choroid plexus papilloma	II	Fourth and lateral ventricles
Choroid plexus carcinoma	III	Lateral ventricles
PINEAL REGION TUMORS		
Pineocytoma	I	Pineal region
Pineal parenchymal tumor of intermediate differentiation	II or III	Pineal region
Pineoblastoma	IV	Pineal region
Papillary tumor of the pineal region	II or III	Pineal region
EMBRYONAL TUMORS		
Medulloblastoma	IV	Fourth ventricle
Atypical teratoid/rhabdoid tumor	IV	Pineal region, suprasellar
Germ cell tumors		Pineal region, suprasellar
MENINGIOMAS		
Meningioma	I	Lateral ventricles
Atypical meningioma	II	Lateral ventricles
Anaplastic meningioma	III	Lateral ventricles
SELLAR REGION TUMORS		
Craniopharyngiomas	I	Suprasellar
Pituitary macroadenoma	I	Suprasellar
OTHER LESIONS AND CYSTS		
Lymphoma		Lateral ventricles
Metastases		All locations
Colloid cyst		Third ventricle
Rathke cyst		Suprasellar
Epidermoid cyst		Fourth ventricle
Dermoid cyst		Suprasellar, pineal
Choroid plexus cyst		Lateral ventricles
Arachnoid cyst		Lateral ventricles

tumors)¹³ or its diminished resorption (e.g., dermoid cysts).¹⁴ Tumor blockage of CSF pathways creates obstructive hydrocephalus, which can be chronic, intermittent, or acute, with the presentation mirroring the time course of CSF flow impediment.

Tumors of the suprasellar region that extend into the third ventricle (e.g., pituitary macroadenoma, craniopharyngioma, optic pathway glioma, Rathke cleft cyst, meningioma) may compress vital structures such as the visual apparatus, pituitary gland, pituitary stalk, or hypothalamus, leading to variable patterns of visual loss or endocrine dysfunction. Disturbance of the hypothalamus may cause metabolic derangements or cognitive and psychiatric disturbances.

Mesencephalic tectal compression may occur in the presence of a pineal region or tectal tumor (e.g., pineoblastoma, tectal

glioma, dermoid cyst, germ cell tumor, meningioma), leading to Parinaud syndrome, which is characterized by up-gaze paralysis, eyelid retraction, convergence-retraction nystagmus, and light-near dissociation. Tumors in this region are prone to causing obstructive hydrocephalus by compression of the narrow sylvian aqueduct.

In the case of fourth ventricular tumors (e.g., juvenile pilocytic astrocytoma, choroid plexus tumors, medulloblastoma, ependymoma) initial symptoms occur because of increased ICP from complete or partially obstructive hydrocephalus. Cerebellar involvement can lead to ataxia or gait disturbance, another frequent presentation of fourth ventricular tumors. Although less common and primarily a late finding, cranial nerve involvement may lead to hoarseness, trouble swallowing, facial weakness, or eye movement abnormalities.

Diagnostic Studies

The wide variety of tumors that originate within or invade the ventricles make it essential for the surgeon to have a systematic method of narrowing the differential diagnosis. Treatment plans for various tumors may differ considerably based on the suspected diagnosis. For example, some germ cell tumors may be diagnosed using only CSF studies and treated effectively with targeted radiotherapy. Similarly, pituitary prolactinomas diagnosed with serum testing often respond dramatically to dopamine agonists. Clues for narrowing the differential diagnosis aside from imaging characteristics include tumor location, patient demographics, and associated conditions.¹⁵ Unique radiographic features of common ventricular pathologies will be discussed later in this chapter.

If surgery is indicated, imaging plays a key role in devising the surgical plan. For example, knowledge of whether the tumor is intraventricular or paraventricular in origin influences the planned approach and goals of surgery. The pursuit of tumor extending outside of the ventricle can result in untoward consequences to neighboring neural structures including cranial nerves, cerebral thalami or hypothalami, internal capsule, and brainstem nuclei. Imaging may also reveal clues as to how a tumor will handle during surgery. Lesions suspected to be particularly firm, vascular, malignant, or invasive may influence the surgeon's choice of access. In addition, advanced imaging sequences may be obtained to further augment planning. For example, constructive interference in steady state (CISS) or fast-spin echo (FSE) turbo techniques (e.g., CUBE, FIESTA, isoFSE, 3D MVOX, SPACE) establish the feasibility of performing an endoscopic third ventriculostomy if deemed necessary. High-resolution T2-weighted imaging may be useful for investigating involved cranial nerves and their direction of displacement. If a feeding or draining vessel appears to be intricately involved, this should be carefully noted and consideration given to dedicated vessel imaging in the form of computed tomographic angiography (CTA), dynamic computed tomographic angiography (dCTA) (Fig. 170.1), or digital subtraction angiography (DSA).¹⁶ Choosing an approach that provides early access and ligation of a feeding arterial pedicle can greatly simplify surgery of vascularized tumors such as meningiomas, choroid plexus tumors, or central neurocytomas.^{4,17} On the other hand, imaging may also be used to rule out a particular approach. For example, a high density of bridging veins draining into the superior sagittal sinus (SSS) may deter the operator from utilizing an interhemispheric approach. Similarly, a low-lying vein of Labbe of the dominant hemisphere may steer the surgeon away from a subtemporal approach to a lesion on that side. We only rarely utilize functional magnetic resonance imaging (fMRI) or diffusion tensor imaging (DTI) because the approaches described in this chapter take advantage of well-established anatomic corridors that minimize the risk to eloquent cortical and subcortical networks.

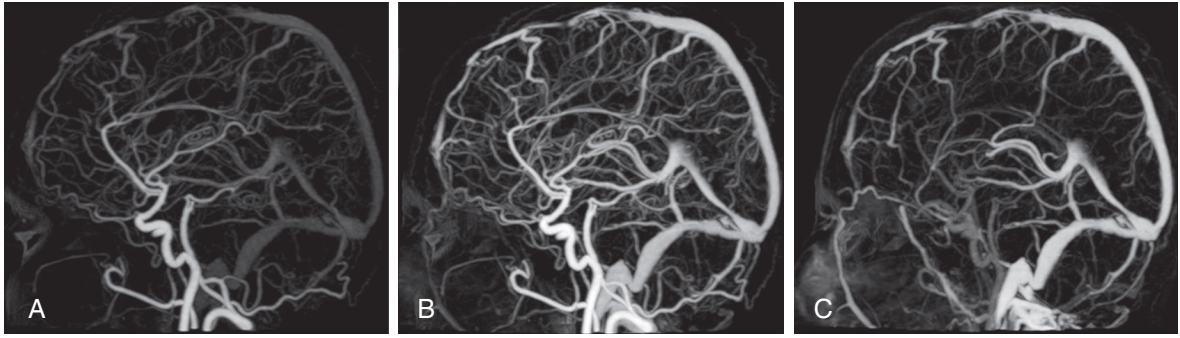


Figure 170.1. Images obtained using dynamic computed tomographic angiography (dCTA) displaying arterial (A), mixed (B), and venous (C) imaging phases. Detailed images from vascular studies such as this aid in decision making and surgical planning.

Surgical Anatomy

Lateral Ventricle

The lateral ventricle is a paired “C-shaped” structure located deep within each cerebral hemisphere wrapping around the diencephalic structures. It has five regions that can be accessed by a variety of surgical approaches. These regions are the anterior (“frontal”) horn, body (“cella media”), atrium (“trigone”), occipital horn, and temporal horn.

Both lateral ventricles communicate with the third ventricle via the foramina of Monro and are bordered by the following structures: the septum pellucidum, thalamus, caudate nucleus, corpus callosum, and fornix. The largest structure in direct contact with the lateral ventricle is the corpus callosum, which overlies the frontal horn and body of the lateral ventricle. It is the largest commissural fiber tract of the brain and is divided into four sections: the rostrum, genu, body, and splenium. Sectioning of the genu or splenium is typically avoided as this may result in disconnection of the two hemispheres. The caudate nucleus makes up the lateral border of both the anterior horn and body of the lateral ventricle as well as part of the roof of the temporal horn. It can be divided into a head, body, and tail. The fornix (Fig. 170.2) connects the hippocampus to the thalamus, hypothalamus, and mammillary bodies, among other limbic structures. It has five main components: the alveus, fimbria, crus, body, and columns. It originates as the alveus running posteriorly as the fimbria, located medial to the hippocampus. The fimbria becomes the crus as it begins to head superiorly, arching superomedially around the posterior thalamus and beneath the splenium before proceeding medial and rostral over the dorsal thalamus to join the crus from the opposite side. This fused midline structure is the body of the fornix, which lies along the medial floor of the lateral ventricles. As the fornix continues rostral toward the foramen of Monro, the fused crura redive to form the columns of the fornix, heading from medial to lateral around the outer margins of the foramen of Monro before heading inferior into the anterior third ventricle to form connections with the anterior commissure, hypothalamus, and mammillary bodies. The posterior border of the foramen of Monro consist of the anterior-superior pole of the thalamus.

The *frontal horn* of the lateral ventricle lies anterior to the foramen of Monro (Fig. 170.3). The medial wall is formed by the septum pellucidum, and the lateral wall is formed by the head of the caudate nucleus. The floor and roof of the anterior horn are established by the rostrum and genu of the corpus callosum, respectively. Of note, there is no choroid plexus within the frontal horn, but it does contain important venous structures that help orient the surgeon. The septal vein runs medially along the septum pellucidum heading toward the foramen of Monro. Although variations exist, the septal vein frequently anastomoses with the thalamostriate vein to form the internal cerebral vein at the level of the foramen of Monro before heading into the velum

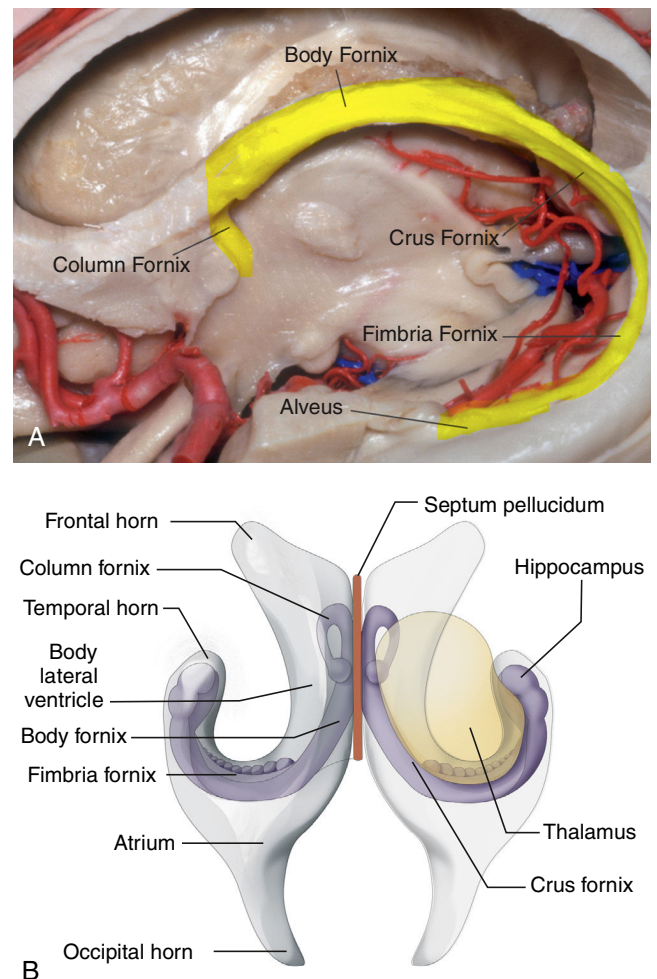


Figure 170.2. (A) Sagittal view of the left fornix (yellow) consisting of the forniceal column, body, crus, fimbria, and alveus, connecting components of the limbic system. (B) A cranial-to-caudal (top down) view of the fornices and their anatomic relationship to the lateral ventricles and thalamus. (A, Courtesy the Rhoton Collection. <https://www.aans.org/en/Education/The-Rhoton-Collection>.)

interpositum (a potential space containing the internal cerebral veins and medial posterior choroidal arteries between the superior and inferior layers of the tela choroidea).^{18,19} (Fig. 170.4) This anastomosis of vessels creates a “U-shape” known as the *venous angle* and can be identified on cerebral angiogram or dedicated CT or MR venography. Typically, this junction is located 3 to 7 mm posterior to the foramen of Monro. Immediately deep (lateral) to

this anastomosis lies the genu of the internal capsule. The septal vein may be safely transected proximal to its junction with the thalamostriate vein to allow for an expanded transforaminal route to the third ventricle.

The *body* of the lateral ventricle extends posteriorly from the foramen of Monro. The lateral wall is formed primarily by the body of the caudate nucleus, the medial wall by the septum pellucidum, the roof by the body of the corpus callosum, and the floor by the thalamus, fornix, choroid plexus, and tela choroidea. The junction between the thalamus and caudate makes up the striothalamic sulcus, which contains the thalamostriate vein. The choroid plexus exists within the choroidal fissure, the intersection between the thalamus and fornix (Fig. 170.5; also see Fig. 170.3). The medial posterior choroidal artery, a branch of the P2 segment, passes lateral to the pineal gland traveling anteriorly within the velum interpositum before heading superiorly through the foramen of Monro and then posteriorly along the choroid

plexus. It is the primary blood supply of the third ventricular and lateral ventricular body choroid plexus and is a common arterial feeder for tumors in these locations.

The atrium makes up the junction between the body of the lateral ventricle and the temporal and occipital horns. The superior and lateral walls are formed by fibers of the corpus callosum, called the *forceps major* and *tapetum* (Fig. 170.6). The tapetum arises from the body and splenium of the corpus callosum and sweeps laterally as it runs toward the occipital lobe to make up the majority of the roof and lateral wall of the atrium. The forceps major is a large bundle of fibers that forms a bulge and makes up the superior portion of the medial border, along with the calcar avis, which is a prominence that overlies the deepest part of the calcarine sulcus. The floor is formed by the collateral trigone, an upward bulge from the posterior end of the collateral sulcus. Anterior to the atrium lies the tail of the caudate, the crus of the fornix, and the pulvinar. The enlargement of the choroid plexus at the trigone is called the *choroid glomus*.

The *occipital horn* extends posterior from the ventricular atrium and is the most variable structure of the lateral ventricle

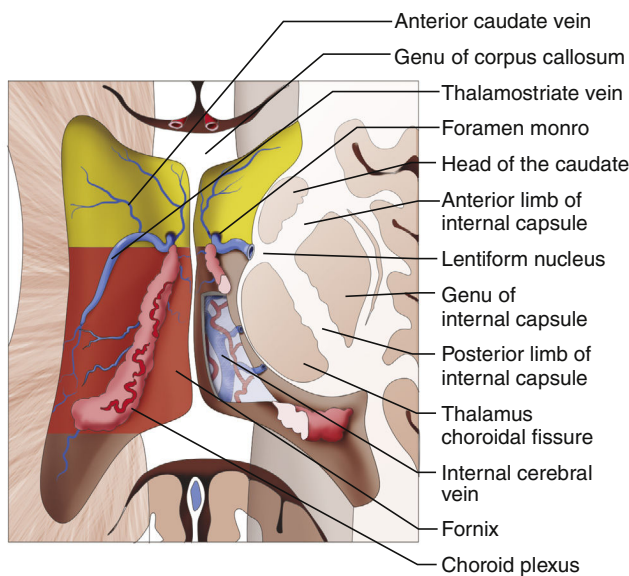


Figure 170.3. Illustrated view of an axial cut across the lateral ventricles revealing the frontal ventricular horns (yellow) and their relationship to the foramen of Monro, caudate nucleus, and corpus callosum. The body of the lateral ventricle is highlighted in red.

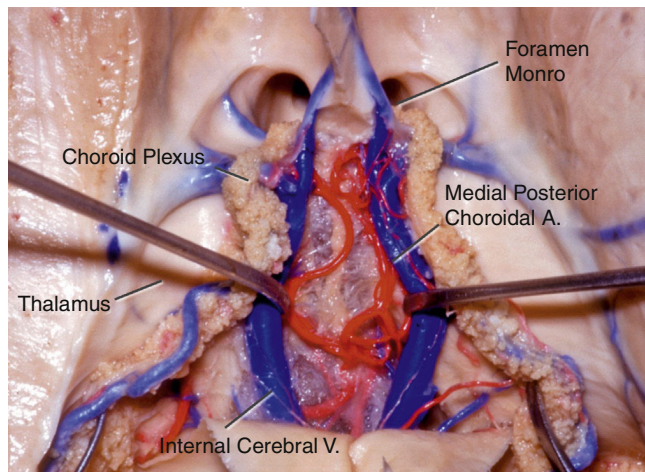
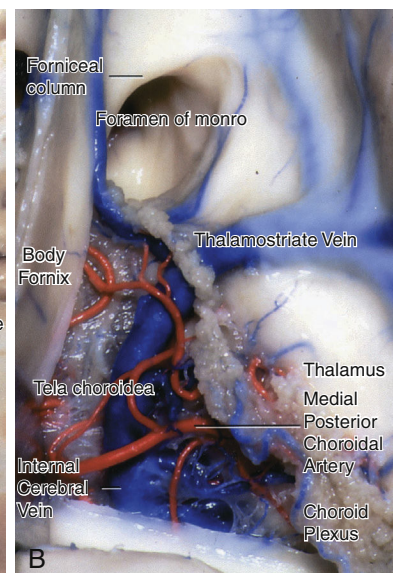
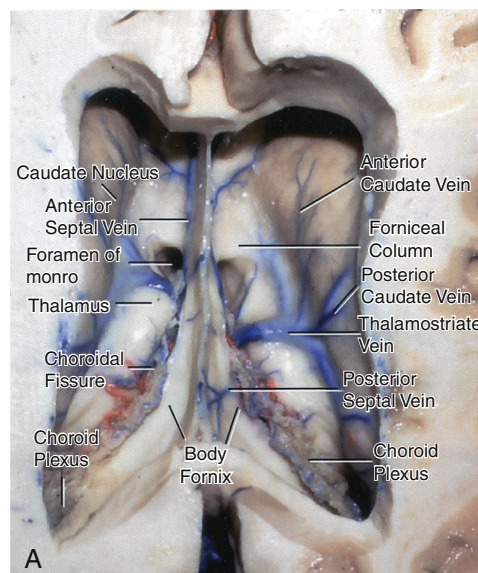


Figure 170.5. Contents of the lateral ventricle body with the intraventricular septum removed and the tela choroidea opened to reveal structures contained within the velum interpositum. A, Artery; V, vein. (Courtesy the Rhoton Collection. <https://www.aans.org/en/Education/The-Rhoton-Collection>.)

Figure 170.4. (A) Cranial-to-caudal view of the relationship of major venous structures within the lateral ventricles. (B) A view of the same venous structures, with the floor of the lateral ventricles opened to reveal the contents of the velum interpositum and their relationship to the vascular anatomy of the lateral ventricle. Note the relationship of the lateral ventricular drainage posterior to the foramen of Monro draining into the internal cerebral vein located within the velum interpositum. (From Rhoton AL. *The lateral and third ventricles*. Neurosurgery. 2002;51[4 Suppl]:S207–S271.)



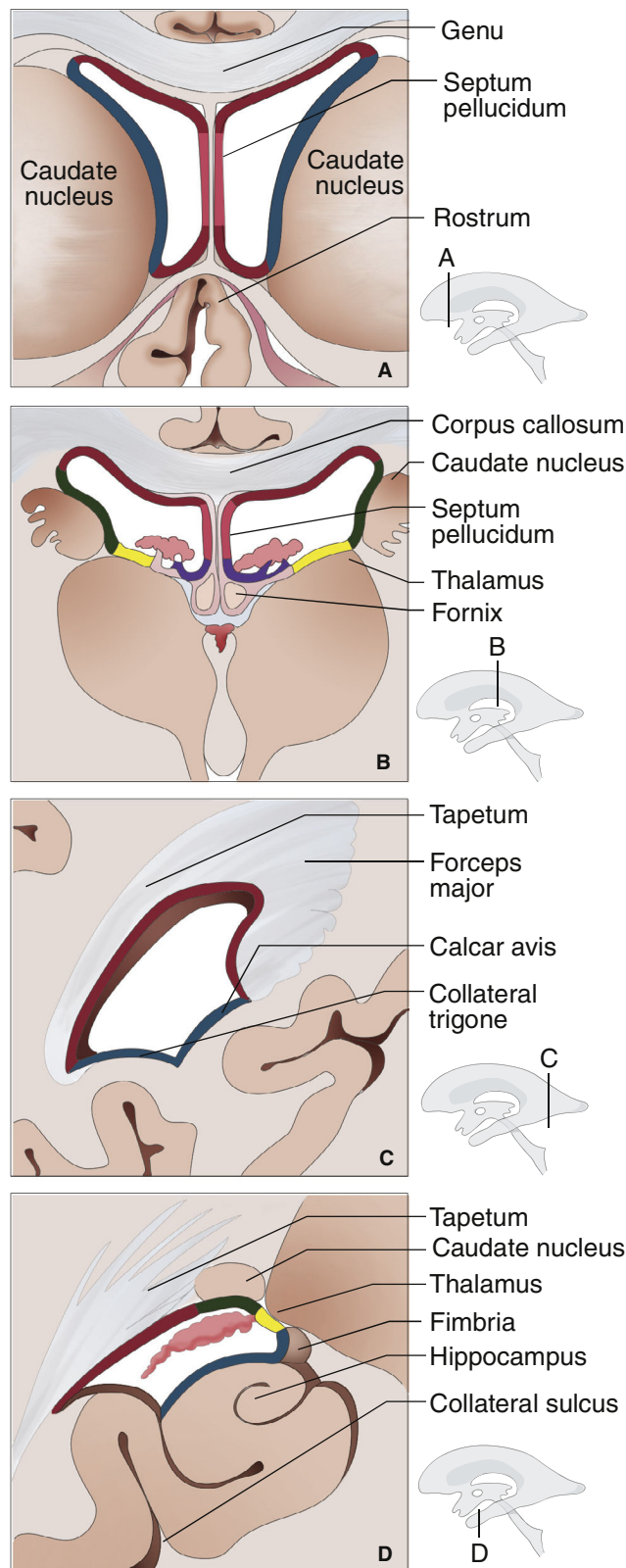


Figure 170.6. Cerebral ventricular anatomy in relation to periventricular structures at various cross-sections. (A) Frontal horn. (B) Body. (C) Occipital horn. (D) Temporal horn. (Modified from Rhoton AL. *The lateral and third ventricles*. *Neurosurgery*. 2002;51[4 Suppl]:S207–S271.)

in terms of its size and shape. As with the atrium, the roof and the lateral wall are formed by the tapetum of the corpus callosum (see Fig. 170.6). The medial border is also formed superiorly by the forceps major and inferiorly by the calcar avis. The floor is defined by the collateral trigone. There is typically no choroid plexus in the occipital horn.

The *temporal horn* begins at the ventricular atrium and ends in the amygdala. The roof is formed medially by the surface of the thalamus and tail of the caudate nucleus, separated from each other by the striothalamic sulcus. Laterally, the roof consists of the tapetum of the corpus callosum, separating the ventricle from the optic radiations (see Fig. 170.6). Along the lateral aspect of the floor, one finds the collateral eminence, which is formed by the underlying deep collateral sulcus and the hippocampal formation located medially. The medial wall superior to the hippocampus is made up of a single structure, the choroidal fissure, which is situated between the thalamus and fornix. This is continuous with the choroidal fissure encountered along the floor of the lateral ventricular body and atrium.

A thorough understanding of the vascular anatomy of the ventricles cannot be overemphasized. As previously mentioned, the choroidal fissure extends posteriorly from the foramen of Monro along the body and atrium of the lateral ventricle turning inferiorly into the temporal horn to terminate at the inferior choroidal point, the entry of the anterior choroidal artery. This artery feeds the choroid plexus of the temporal horn and part of the atrium. The medial posterior choroidal artery, discussed previously, supplies the choroid plexus of the third ventricular roof along with that of the body and atrium of the lateral ventricle. The majority of lateral ventricular tumors derive their arterial supply from branches of these choroidal arteries.

The veins of the lateral ventricles typically drain into either the internal cerebral veins or basal veins of Rosenthal. Noteworthy vessels include the anterior septal and caudate veins, thalamostriate vein, superior choroidal vein, and medial atrial vein (see Figs. 170.4 and 170.5). Together, these either anastomose with each other or drain directly into the internal cerebral vein at the posterior margin of the foramen of Monro as it transitions into the third ventricle. The inferior choroidal vein, inferior ventricular vein, lateral atrial vein, and the veins of the amygdala and hippocampus drain into the basal vein of Rosenthal. Sacrifice of the internal cerebral or basal vein of Rosenthal is not advised. Regardless, veins should be spared whenever possible to avoid the possibility of venous infarct or hemorrhage.

Third Ventricle

The third ventricle connects anterior-superiorly to the lateral ventricles via the paired foramina of Monro, and the fourth ventricle connects posterior-inferiorly via the sylvian aqueduct. The lateral wall is formed by the thalamus and hypothalamus, which are separated from each other by the hypothalamic sulcus. In approximately 75% of patients, the *massa intermedia* spans the third ventricle, connecting the two thalami.

The floor of the third ventricle, which extends from the optic chiasm to the sylvian aqueduct, includes the infundibular recess, tuber cinereum, mammillary bodies, and posterior perforated substance (Fig. 170.7A). The roof of the third ventricle extends from the foramen of Monro to the suprapineal recess and is formed by four layers that proceed from superior (dorsal) to inferior (ventral): the fornix, the two layers of the tela choroidea (between which exists the velum interpositum), and the choroid plexus (see Figs. 170.5 and 170.7A). The velum interpositum contains the internal cerebral veins, which drain posteriorly to the vein of Galen, and the medial posterior choroidal arteries, which originate posteriorly and course anteriorly toward the foramen of

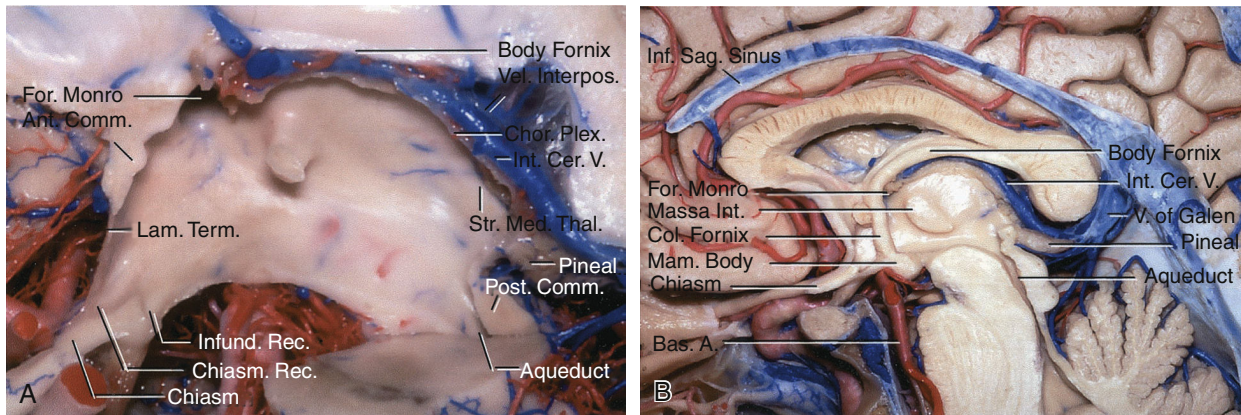


Figure 170.7. Midsagittal views of the third ventricle. (A) Enlarged view to better demonstrate intraventricular contents. (B) Global view to demonstrate the third ventricle's relationship to surrounding structures. (From Rhoton AL. *The lateral and third ventricles*. *Neurosurgery*. 2002;51[4 Suppl]:S207–S271.)

Monro. The two internal cerebral veins run close to each other up to the pineal recess, where they deviate from the midline and proceed along the superolateral surface of the pineal body to the deepest point of the splenium of the corpus callosum to form the vein of Galen. The choroid plexus of the third ventricle is attached to the inferiormost layer of the tela choroidea, wrapping anteriorly and superiorly through the foramen of Monro into the lateral ventricles.

The anterior border of the third ventricle extends from the optic chiasm to the foramen of Monro and is formed by the optic chiasm, lamina terminalis, anterior commissure, and the column of the fornix (see Fig. 170.7A). The lamina terminalis is a thin sheet of gray matter and pia that attaches from the superior aspect of the optic chiasm to connect to the rostrum of the corpus callosum.

The posterior border of the third ventricle extends from the sylvian aqueduct to the suprapineal recess. Between these structures are the posterior commissure, pineal body, pineal recess, and the habenular commissure located between the crura of the fornices anterior to the pineal body (see Fig. 170.7B).

Anatomically, tumors affecting the third ventricle originate from any of four primary regions: (1) suprasellar with expansion into the ventricle (e.g., craniopharyngiomas, pituitary adenomas, meningiomas, optic pathway gliomas), (2) the ventricular wall (e.g., ependymomas, hypothalamic or thalamic gliomas), (3) intraventricular structures (e.g., choroid plexus tumors, meningiomas), and (4) the pineal region (e.g., pineal gland tumors, germ cell tumors, meningiomas, tectal tumors).

Fourth Ventricle

The fourth ventricle appears to be rhomboid when viewed from a dorsal approach during surgery (Fig. 170.8). It is contained by the cerebellum posteriorly, the brainstem anteriorly, and the cerebellar peduncles laterally. In the craniocaudal direction, the fourth ventricle extends from the sylvian aqueduct to the obex. The ventricular floor (see Fig. 170.8A) is on its ventral surface, formed by the dorsal surface of the pons and the medulla; its lateral wall is formed by the superior, medial, and inferior cerebellar peduncles. The ventricular roof appears tent-like when viewed directly lateral on sagittal imaging. Describing it from rostral to caudal, it expands from a narrow point at its rostral end in both lateral and dorsal dimensions to form a “peaked” apex about halfway between the aqueduct and obex. This apex is its area of greatest dorsal height and lateral width. As it continues caudal toward the obex, it heads ventrally and tapers medially to a point at the level of the foramen of Magendie. The triangular area rostral to the apex is the superior aspect of the roof. The triangle caudal to the apex is the inferior aspect of the roof.

The ventricular-facing surface of the superior roof can be divided into median and lateral areas. The median portion consists of the superior medullary velum. Dorsal to the superior medullary velum is the superior cerebellar vermis, which becomes the fastigium at the ventricular apex. The lateral walls of the ventricle are formed by the lingula of the superior cerebellar peduncles and rostral portion of the middle cerebellar peduncles. The inferior roof slopes ventrally as its heads caudally, away from the apex and fastigium, to its attachment at the inferolateral borders of the ventricular floor. The inferior medullary velum continues caudally from the superior medullary velum beginning at the level of the fastigium. Dorsal to this thin sheet of neural tissue is the nodule of the vermis. Further caudally along the inferior roof, the inferior medullary velum attaches to the tela choroidea. The tela choroidea forms the caudalmost aspect of the inferior roof as well as its lateral recesses. It is partially covered by the uvula of the vermis along its dorsal aspect. The junction between the inferior medullary velum and the tela choroidea runs inferolaterally from the vermis nodule to the lateral recesses. As with the tela choroidea of the third ventricle, it contains a vascular layer between its leaves and is the origin of the fourth ventricular choroid plexus. The tela choroidea is fixed to the inferolateral margins of the ventricular floor by the taenia of the fourth ventricle (see Fig. 170.8D–E).

To obtain a wide view into the fourth ventricle, the lower structures of the vermis must be elevated and retracted dorsally and superiorly. Retraction must be gentle to avoid injuring the vermis. For this reason, adequate arachnoid dissection and sectioning of the tela choroidea and inferior medullary velum are necessary to release tension on these structures. In practice, the telovelar approach provides excellent and safe access to the fourth ventricle.

The posterior inferior cerebellar artery (PICA) is the most important artery related to the fourth ventricle. Its course can be divided into five segments: the anterior medullary segment, lateral medullary segment, tonsillomedullary or posterior medullary segment, telovelotonsillar or supratonsillar segment, and distal segment (Fig. 170.9). Primarily, the first three segments are the origin of the arterial supply to the lower brainstem and vermis, but these segments may also supply tumors located in the fourth ventricle. Therefore, it is particularly important to study the individual course of the PICA and to distinguish between branches of the PICA that exclusively supply the tumor and those that supply the brainstem and cerebellum.

Indications for Surgery

The decision to operate on a ventricular tumor depends on the suspected pathology and patient presentation. For example, most

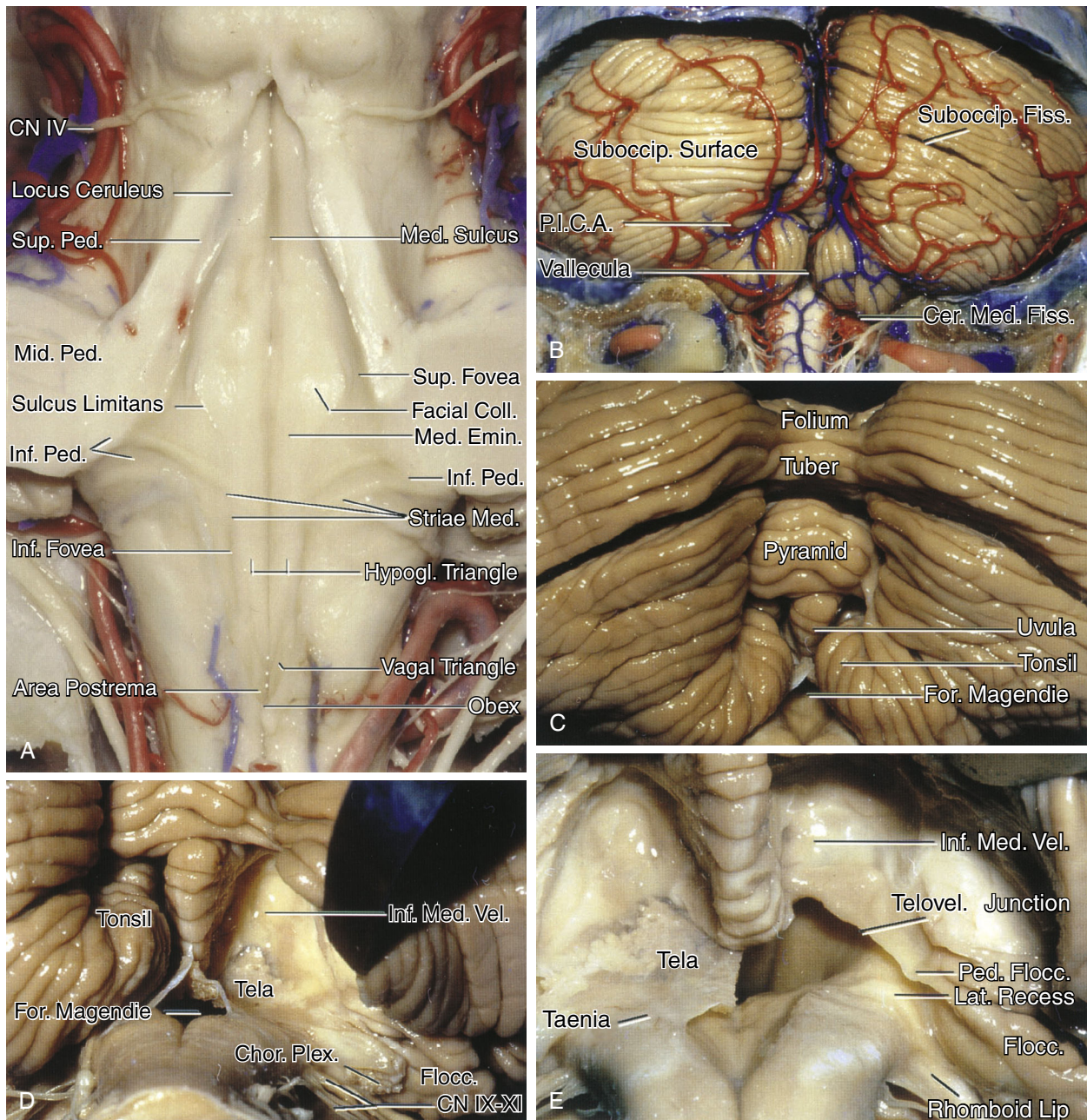


Figure 170.8. (A) Dorsal view of the fourth ventricular floor following sectioning of the cerebral peduncles and removal of dorsal elements. (B) Dorsal view of the cerebellar surface. (C) Enlarged midline view of midline cerebellar structures. (D) Enlarged midline view of midline cerebellar structures showing the right cerebellar tonsil retracted to expose the lower part of the fourth ventricular roof formed by the inferior medullary velum. (E) Enlarged view of midline cerebellar structures with the bilateral tonsils and right tela choroidea removed to expose the fourth ventricle cavity and floor. (From Rhoton AL. *Cerebellum and fourth ventricle*. *Neurosurgery*. 2000;47[3 Suppl]:S7–S27.)

practitioners will expectantly manage an asymptomatic colloid cyst less than 1 cm in diameter or a nonobstructive benign-appearing pineal cyst.^{20–23} Nonsurgical management of prolactinomas or germinomas must be considered because of the efficacy of noninvasive treatments.

In addition, when deciding on surgery, practitioners must account for the patient's overall health and the mechanism by which the pathology is causing the patient's symptoms. A familiar scenario encountered by clinicians is a high-risk surgical lesion in an older adult or medically frail patient suffering from obstructive hydrocephalus. In such instances, CSF diversion through an endoscopic ventriculostomy or shunt placement procedure may be preferable to directly operating on the lesion.

Ultimately, for cases requiring definitive diagnosis or symptom management, surgery often plays an essential role in patient management.

Surgical Planning

Because of the rare occurrence of ventricular tumors, their deep location within the cerebrum, and difficult access to their vascular supply, errors can have devastating consequences. The successful execution of ventricular tumor surgery requires thoughtful preoperative decision making, selection of appropriate operating room resources, and close coordination with operating room personnel.

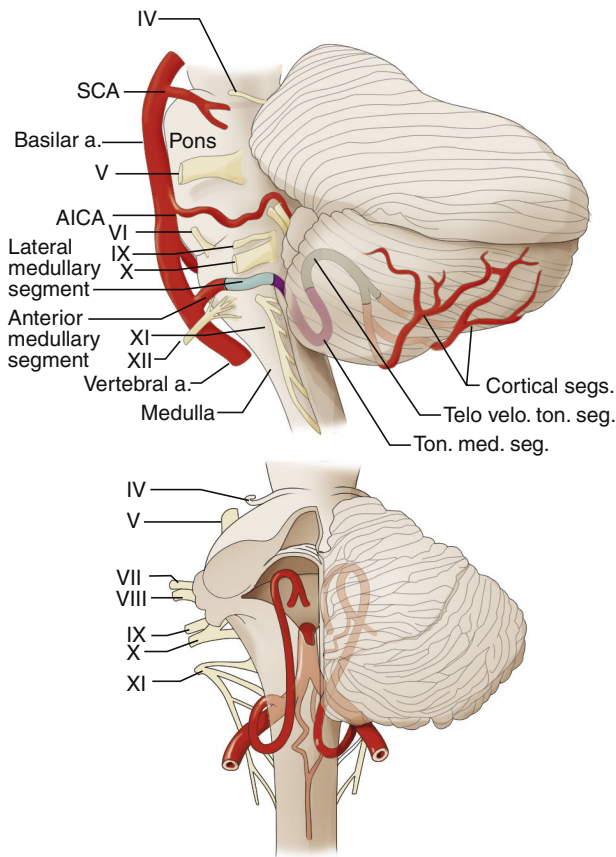


Figure 170.9. Lateral view of the brainstem and cerebellum depicting the course of the posterior inferior cerebellar artery and its five segments: anterior medullary, lateral medullary, tonsillomedullary, telovelotonsillar, and distal segments. AICA, Anterior inferior cerebellar artery; SCA, superior cerebellar artery. (Modified from Fukuda H, Evins AI, Iwasaki K, et al. The role of alternative anastomosis sites in occipital artery-posterior inferior cerebellar artery bypass in the absence of the caudal loop using the far-lateral approach. *J Neurosurg*. 2017;126[2]:634–644.)

After obtaining prerequisite laboratory studies (e.g., endocrinologic and/or CSF studies), advanced imaging to assess cortical function, subcortical pathways, CSF flow dynamics, and cranial nerve or vascular involvement should be selectively obtained with careful thought to the planned surgical approach. For example, if a transparietal or transtemporal approach to an atrial lesion is anticipated, fMRI and tractography may be useful to assess the relationship of cortical and subcortical areas at risk. Likewise, vascular anatomy should be examined judiciously before committing to a surgical approach. A high density of superficial bridging veins to the SSS may favor a transcortical approach over an interhemispheric approach. A low-riding vein of Labbe crossing the dominant temporal lobe can be a deterrent for utilizing a subtemporal approach because of the risk for venous infarction resulting in language dysfunction. Inferior displacement of the vein of Galen by a pineal region tumor increases the challenge and risk for the supracerebellar infratentorial approach. Medially located arterial pedicles to a vascular tumor may favor an approach allowing for their early exposure and ligation.

Once an operative approach has been selected, patient positioning is key to capitalizing on its benefits. This is especially

important to maximize tumor exposure and utilize gravity after initial debulking is completed, while avoiding surgeon frustration and fatigue. Commonly used patient positions for ventricular tumors include: supine, three-quarters prone/park-bench, prone, Concorde, and sitting positions. We use a cardiac Doppler imaging and monitor end-tidal PCO₂ when using the sitting position to allow for detection of air embolus.

Planning and preparing the operative field for emergency external ventricular drain placement may help salvage a patient in an emergency. If a tumor suddenly hemorrhages or a large territory infarct occurs, the compartmental ICP may increase dramatically and lead to cerebral herniation. Having a ventricular drain in place or readily deployable allows for rapid ICP control, affording the operator time to perform more definitive ICP-lowering maneuvers. If the patient initially presented with communicating hydrocephalus, we may leave an external ventricular drain postoperatively to minimize the risk for CSF leak and maximize the opportunity for successful dural closure. A graft or dural substitute should be used if needed to obtain a watertight dural closure. Subsequently, the Valsalva maneuver confirms the integrity of the closure.

Planning for concurrent surgical treatments is also important when considering how to best handle ventricular tumors. For example, in the presence of obstructive hydrocephalus, endoscopic CSF diversion may be performed during the same procedure for tumor biopsy or removal. If this decision is made, the surgeon may elect to perform an endoscopic third ventriculostomy before tumor manipulation as subsequent bleeding may significantly obscure the endoscopic field and prevent safe or adequate fenestration of the floor of the third ventricle. Similarly, if the tumor is expected to be aggressive and early chemoradiation is crucial to tumor control, a surgical approach allowing rapid wound healing may be selected over creation of an extensive scalp wound that will delay adjuvant treatment.

Surgical Adjuncts

Although there is no substitute for anatomic knowledge, microsurgical experience, and meticulous planning, surgical adjuncts should be thoughtfully employed to facilitate safe removal of ventricular tumors. In addition to maintaining the proper operative trajectory and avoiding surface vessels, neuronavigation helps in planning the skin incision and locating the craniotomy. It is also exceptionally useful for determining the initial boundaries of the tumor, but it can be limited by its reliance on static preoperative images. Intraoperative ultrasonography is useful for localization of intraventricular lesions, planning a corticectomy, and confirming in real-time the operative trajectory and extent of resection.

The use of mannitol, furosemide, or lumbar CSF drainage may facilitate arachnoid dissection in select cases, especially when the brain is swollen. However, we do not routinely employ these techniques as they are often not necessary for achieving adequate access to the tumor and place the patient at risk for complications such as brainstem herniation, remote cerebellar hemorrhage, and subdural hematoma formation. We prefer to plan our surgical approach to achieve early access of the ventricles or basal cisterns for CSF release and cerebral relaxation.

Given the myriad important deep brain and neurovascular structures associated with ventricular tumors, we utilize neuromonitoring during all cases of ventricular tumor surgery. Somatosensory evoked potentials help the surgeon detect reversible impending issues such as ischemia caused by inadvertent overretraction of the cerebrum. The use of continuous brainstem auditory evoked responses (BAERs) provides similar information regarding the integrity of brainstem pathways and can be especially useful during fourth ventricular tumor surgery. Motor evoked potentials (MEPs) can be conducted to perform interval assessment of the pyramidal tracts, while electromyography

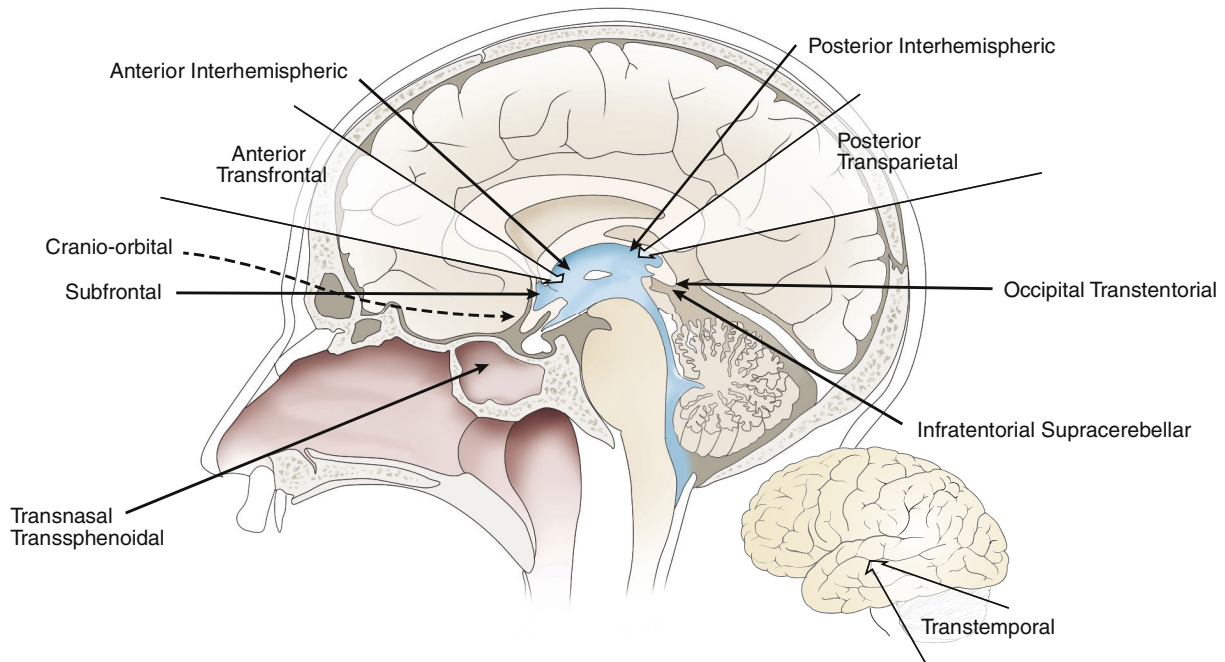


Figure 170.10. Artist's depiction summarizing the major approaches to the ventricular and periventricular regions.

(EMG) can be used to identify abnormally located cranial nerves entrapped within or displaced by tumor.

The use of the intraoperative MRI has been touted for its ability to increase the degree of tumor resection in select cases.^{24,25} This may reduce the need for reoperation or postoperative radiotherapy for residual tumor. However, given the advances in endoscopic surgery and limitations introduced by MRI-compatible operating rooms, this tool is used in select cases.

Technologic advancements since Dandy's ventriculoscope have made the modern endoscope smaller, lighter, more maneuverable, and capable of generating high-resolution video. Endoscopes may be used during any portion of the surgery but have demonstrated exceptional value when used with an angled or flexible tip to inspect postoperative cavities for residual tumor, overcoming the line-of-sight limitation of the operative microscope.

SURGERY OF THE LATERAL VENTRICLE

Principles

Access to the tumors within the deep-seated ventricular system requires transgression of cortical and/or subcortical structures.^{11,26,27} Approaches to the lateral ventricles aim to minimize parenchymal transgression, while allowing for adequate access to the tumor through the potential spaces of the brain (i.e., sulci, fissures, and cisterns). Described in 1921 by Dandy,^{3,28} the interhemispheric transcallosal approach has become a favored method for accessing lateral ventricular tumors because of its minimal cortical disruption, avoidance of lobar projection and association fiber transgression, and a theoretically reduced risk for postoperative seizures (although this last point remains contested in the literature).^{27,29–32} The major risks associated with transcallosal surgery lie in the possibility of inducing disconnection between the two cerebral hemispheres, most commonly with damage to the genu or splenium of the corpus callosum, or abulia from injury to the cingulate gyri.^{26,27,30,33}

In our experience, approaching the tumor along its long axis allows the operator to utilize a narrower operative window, obviating the need for unnecessary resection and retraction of

cerebral tissue. Once inside the lateral ventricle, the operator should take advantage of the rich intraventricular anatomy for orientation including the foramen of Monro, septum pellucidum, choroid plexus, thalamostriate, and the anterior septal and caudate veins.

We will proceed by introducing the most common surgical approaches to the lateral ventricles based on their ability to access pathology within the following sites: frontal horn, ventricular body, atrium/trigone, temporal horn, or occipital horn (Fig. 170.10). Table 170.2 provides a framework for categorizing the various approaches and understanding the advantages and shortcomings of each.

Surgical Approaches

Anterior Interhemispheric Transcallosal Approach

The anterior interhemispheric transcallosal approach to the lateral ventricle provides ample access to the frontal horn and body of the lateral ventricle, and it is adequate treatment for most pathology encountered in these regions. Its main limitation exists in its ability to treat large tumors with significant lateral extension or bilateral ventricular involvement. This approach takes advantage of the potential space between the falx cerebri and the mesial frontal cortical structures encountered en route to the corpus callosum, namely the superior frontal gyrus, paracentral lobule, and cingulate gyrus, obviating the need for corticectomy.

Careful study of the SSS, which is frequently lateral to the midline,^{34,35} and the density of draining veins determines the feasibility of this approach. In general, there exists a lesser density of draining veins at the anterior aspect of the SSS.³⁶ Despite a common belief that veins draining to the anterior portions of the SSS as well as the anterior sinus itself can be readily sacrificed, we make every effort to spare these vessels to avoid the potential for venous infarction.^{36,37}

Positioning of the patient depends on the surgeon's preference. The patient may either be placed in the supine position with the neck flexed and the midline axis directly perpendicular to the floor or lateral with the vertex slightly up toward the ceiling and the midline parallel to the floor. The lateral position can be helpful

TABLE 170.2 Lateral Ventricular Approaches

Category	Approach ^a	Ventricular Access				Advantages	Disadvantages	Risks
		Frontal Horn	Body	Atrium	Temporal Horn	Occipital Horn		
Midline	Anterior Interhemispheric Posterior Interhemispheric	Transcallosal	☒			Low cortical vein density Avoids cortical transgression Early access to medial posterior choroïdal artery vascular pedicles Avoids cortical transgression	Limited lateral view and operative angles Limited lateral view and operative angles Approach may be limited by cortical vein density	Abulia with bilateral cingulate injury May result in disconnection syndrome Risk to medial visual cortex, especially with excessive retraction
		Transcallosal	☒					
		Transcingulate and transprecuneus	☒	☒	☒	Expanded lateral working angles compared with transcallosal approach Early access to medial posterior choroïdal artery pedicles through superomedial approach Reduced risk to optic radiations, white matter tracts, and functional cortices compared with lateral approaches	Narrow operative corridor Limited access to occipital horn Limited by cortical vein density	May result in disconnection syndrome Risk to medial visual cortex, especially with excessive retraction
Lateral	Supracerebellar Transtentorial	Trans-collateral sulcus	☒	☒	☒	Early access to medial posterior choroïdal artery pedicles through inferior approach Saves eloquent cortices and fibers of the optic radiation Early access to anterior choroïdal artery pedicles Improved lateral access to ipsilateral frontal horn and contralateral frontal horn compared with midline approaches Avoids midline draining veins Improved lateral access to ipsilateral lesions including access to paraventricular lesions	Long operative reach Narrow operative corridor with constrained working area Limitations to degree of contralateral reach	Supplemental motor area syndrome Mutism, abulia, memory disturbance, seizures
	Transfrontal	Trans-middle frontal gyrus or superior frontal sulcus	☒	☒				
Lateral	Transparietal	Trans-superior parietal lobule or intraparietal sulcus	☒	☒			Late access to inferomedially located posterior choroïdal artery pedicles Risk to functional cortices and white matter tracts	Gerstmann syndrome/acalculia, finger agnosia, agraphia, left-right disorientation (if dominant side) Apraxia on either side Receptive aphasia on dominant side (risk to Wernicke area) Contralateral superior quadrantanopia Contralateral superior loop
	Anterior Neocortical Resection	Temporal lobectomy		☒		Decreased risk to receptive language on dominant side	Limited posterior temporal reach	Contralateral superior loop

Category	Approach ^a	Ventricular Access				Advantages	Disadvantages	Risks
		Frontal Horn	Body	Atrium	Temporal Horn	Occipital Horn		
Transtemporal	Trans-middle or inferior temporal gyrus				☑		Early access to anterior choroidal artery pedicles Short operative reach	Receptive aphasia on dominant side (risk to Wernicke area) Contralateral superior quadrantanopia (risk to Meyer loop)
Subtemporal	Posterior subtemporal approach or trans-occipitotemporal sulcus			☑	☑	☑	Inferior temporal gyrus or occipitotemporal sulcal entry avoids Meyer loop and Wernicke area Early access to anterior choroidal artery pedicles	May require significant temporal lobe retraction leading to venous infarct or cortical ischemia CSF leak if temporal or mastoid air cells are encountered and not closed
Transylvian	Transylvian fissure				☑		Avoids Meyer loop and Wernicke area Access to paraventricular tumors of the mesial temporal lobe Spare neocortical structures Direct access to occipital horn	Vascular injury (middle cerebral artery) Expressive aphasia on dominant side
Transoccipital	Trans-superior or middle occipital gyrus; Trans-parietooccipital or occipital sulcus					☑	May require transgression of visual fibers Late access to choroidal artery pedicles	Homonymous hemianopia; therefore mostly reserved for patients with severe and unrecoverable preoperative visual deficit

^aBolded approaches represent commonly used “workhorse” approaches.

in its exploitation of gravity to help retract the lower cerebral hemisphere away from the falx cerebri.

Depending on the location of the tumor, the incision and craniotomy are planned. Typically a horseshoe flap with a midline pedicle, abbreviated bicoronal incision, or straight linear incision is used. Navigation confirms the surgical trajectory and location of the SSS and draining veins. This helps plan the final craniotomy which, for frontal horn or ventricular body lesions, should not need to be located more than 3 to 4 cm posterior of the coronal suture to avoid injury to the motor cortex. Two burr holes are placed on either side of the SSS to allow for its complete exposure. The final size of the craniotomy depends on the patient's anatomy and expected pathology. If during epidural dissection the dura of the SSS is tenaciously adherent, a two-piece craniotomy may also be considered to decrease the risk for damaging the sinus.

Upon exposure of the dura and SSS, the operating microscope is brought in and the dura is opened in a C shape with the pedicle toward the sinus. Care is taken when opening the dura to protect and spare any draining veins. Sharp microdissection of the veins from the leptomeninges or dura is utilized to allow for adequate dural reflection without unnecessarily damaging bridging veins. Sutures are placed in the superior aspect of the falx, right below the SSS to rotate the sinus gently away from the operative corridor and maximize the surgeon's view. However, care must be taken to avoid placing excessive tension on, or occluding, the SSS, which could lead to its thrombosis.

The operation is continued down the interhemispheric corridor using sharp microdissection to remove any arachnoid bands that are encountered. This helps relieve tension, allowing the cerebral hemisphere to fall away from the falx. Releasing additional arachnoid bands at the ends of the dissection allows for additional hemispheric retraction and wider exposure. To help maintain the operative corridor, we place surgical cotton balls into the anterior- and posterior-most recesses for gentle retraction. Flat patties may be used to protect the mesial cortices of the brain. We do not use retractors because of the possibility of direct parenchymal or vascular injury and the slightly more restrictive view of the operative working space that occurs when the retractor blades are established.

The ipsilateral callosal marginal artery is often identified before encountering the cingulate gyrus and should not be mistaken for the deeper pericallosal artery. The callosal marginal artery dives between the two cingulate gyri, helping the surgeon identify the midline to begin separating the adjoining cinguli and avoid subpial injury. One may follow this artery along its proximal course toward the A2 bifurcation to identify the pericallosal artery, which can be used for anatomic orientation and surgical plane definition. On occasion, the cingulate gyri may adhere to each other at a point immediately deep to the site where the falx ends, just above the corpus callosum. The operator must be sure not to confuse adherent cinguli as the corpus callosum. It is helpful to note that gyri lack the classic white "sheen" of the corpus callosum.³⁸ After identifying the pericallosal arteries, they can be followed posteriorly, with the corpus callosum lying beneath. The corpus callosum should be widely exposed, and the paired pericallosal arteries should be identified. Once this anatomic view is established, a corpus callosotomy may be performed posterior to the genu. The callosal midline is often demarcated by a small callosal artery, which can help the operator avoid entering the contralateral ventricle. The location of the intended callosotomy should be confirmed with navigation, and an adequate opening made. A 2-cm callosotomy is adequate for most intraventricular pathology.

Standard anatomy should be confirmed upon entering the ventricle. Particularly important structures for orienting the surgeon include: (1) the septal vein, (2) thalamostriate vein, (3) choroid plexus, and (4) foramen of Monro. The septal vein

exists medial to the foramen of Monro, while the thalamostriate vein is lateral. If additional relaxation of the ventricular cavity is required, the septum pellucidum may be fenestrated to allow for an alternate route for CSF egress.

During tumor resection, special attention must be paid to the point at which the thalamostriate vein turns medially toward the foramen of Monro. At this point, the genu of the internal capsule exists just deep within the lateral ventricular wall.^{19,39,40} Upon tumor removal, we routinely employ an angled endoscope to confirm satisfactory tumor resection before closure.

Posterior Interhemispheric Transcallosal Approach

The posterior interhemispheric transcallosal approach provides access to tumors of the posterior ventricular body and anterior atrium of the lateral ventricle as well as the third ventricle.⁴¹ It shares many similarities and technical challenges with the previously described anterior interhemispheric transcallosal approach.

Like the anterior interhemispheric approach, a thorough preoperative assessment should take place before selecting this operation. Knowledge of venous anatomy is once again key to ensuring feasibility of this approach. The density of surface veins draining to the SSS is greater in the posterior frontal and parietal lobes, becoming most sparse around the occipital lobes.^{42,43} Appraisal of the patient's preoperative visual status is paramount given the risk for injury to the visual cortex of the medial occipital lobe or splenium. In the case of homonymous hemianopsia in the visual field contralateral to the dominant hemisphere, extensive damage to the splenium of the corpus callosum would interrupt visual information from the nondominant hemisphere from reaching the speech networks located primarily on the dominant hemisphere leading to alexia without agraphia—a debilitating disconnection syndrome.^{44,45}

The patient may once again be positioned in the supine or lateral position, with an even greater degree of neck flexion than that used for the anterior interhemispheric approach. Some surgeons choose to perform the operation with the patient prone. A craniotomy centered over the parieto-occipital region, created in a similar fashion to that used in the anterior interhemispheric approach, is planned using navigation. The same steps are taken to avoid damaging draining veins or the SSS, although there are fewer draining veins located in this region. Dissection along the posterior interhemispheric fissure takes place along the banks of the precuneus and cingulate gyrus. Coming from this posterior angle, only the pericallosal artery will be visualized, as the callosal marginal artery runs superior and branches into distal vessels in the region of the central sulcus. A callosotomy is performed in the posterior body of the corpus callosum without damage to the splenium, allowing access to the posterior compartments of the lateral ventricular system.

Posterior Interhemispheric Transcingulate and Transprecuneus Approach

A variation of the posterior interhemispheric approach, this technique allows increased access to the lateral aspect of the ventricular body or atrium while requiring only minimal cortical transgression.^{11,46,47} In addition, it allows for ready access to the medial posterior choroidal arterial pedicles for early tumor devascularization. Traditional lateral transcortical approaches to the atrium via the temporal or parietal lobe risk damaging overlying functional cortices, subcortical tracts, or optic radiations as they travel lateral to the atrium and occipital horn.⁴⁸ The interhemispheric approach avoids direct risk to these structures.¹⁹

Rather than performing a callosotomy as described previously using the posterior interhemispheric transcallosal approach, an opening can be made within the precuneus or cingulate isthmus to access the ventricular atrium.⁴⁶ Some surgeons elect to use a

contralateral craniotomy with a transfalcine window to further increase the lateral view and attain a more natural working angle.⁴⁹ If working through the contralateral corridor, the operator must take exceptional care not to overretract the occipital lobe.^{50,51} In this scenario, the use of gravity and dynamic retraction is generally favored.

Transfrontal Approach

The transfrontal approach to the frontal horn, lateral ventricular body, or third ventricle can be performed using either a transcortical (Video 170.2) or transsulcal technique. The transfrontal approach is best utilized in patients with hydrocephalus, which presents a wide operative corridor for the surgeon (Fig. 170.11). It is especially useful when an unacceptably high density of midline bridging veins prevents safe dissection of the interhemispheric fissure or if the tumor contains a significant lateral component that would be difficult to reach using an interhemispheric approach.

The patient is placed in the supine position with head flexion and a slight turn. A linear incision is planned using navigation followed by a simple craniotomy exposing the superior and middle temporal gyri. Either a middle frontal gyrus corticectomy or superior frontal sulcus dissection is performed. The dominant-sided superior frontal gyrus houses the supplementary motor area and should not be used as a site for corticectomy. Navigation confirms the proper trajectory.

The main limitations of this approach are the lack of anterior-posterior and contralateral reach, particularly if the tumor extends superiorly into the contralateral frontal horn. Care must be taken not to cause retraction injury or venous infarction, which could injure the supplemental motor area and result in motor apraxia. Tumors of the middle or posterior ventricular body requiring a more posterior corticectomy are at increased risk for damaging

the motor cortex.⁴⁶ Reportedly the most common complications of the transfrontal approach in order of incidence are seizure, transient mutism, hemiparesis, short-term memory disturbance, abulia, and attention deficit.⁵²⁻⁵⁴

Transparietal Approach

The transparietal approach via the superior parietal lobule or intraparietal sulcus is well described.^{10,26,52,55,56} It provides reliable access to both the medial and lateral aspects of the ventricular atrium as well as the posterior temporal and anterior occipital horns.^{27,52,56,57} This approach should be considered for large atrial masses or tumors with significant lateral paraventricular extension that are not amenable to a midline interhemispheric approach.

Navigation is used to plan the appropriate incision and craniotomy. Local surface anatomy and ultrasound guidance should be used to confirm the accuracy of navigation before initiating corticectomy or sulcal dissection toward the ventricular atrium. Late access to choroidal arterial feeders positioned inferomedial beneath an atrial tumor may dissuade the surgeon from using this approach if the tumor is expected to be highly vascular.^{58,59}

Knowledge of regional anatomy is key to avoiding complications with this approach. The optic radiations can be found coursing along the lateral walls of the atrium. Tumors adjacent and adherent to the lateral atrial wall should be debulked before removal from the lateral ependymal surface to avoid excessive and undue traction of the optic radiations at this level. Attendant risks associated with this approach include the possibility of causing contralateral inferior quadrantanopia, aphasia, neglect, or Gerstmann syndrome.^{26,48} Aphasia and Gerstmann syndrome are primarily complications that arise following damage to the angular or supramarginal gyri of the dominant hemisphere.

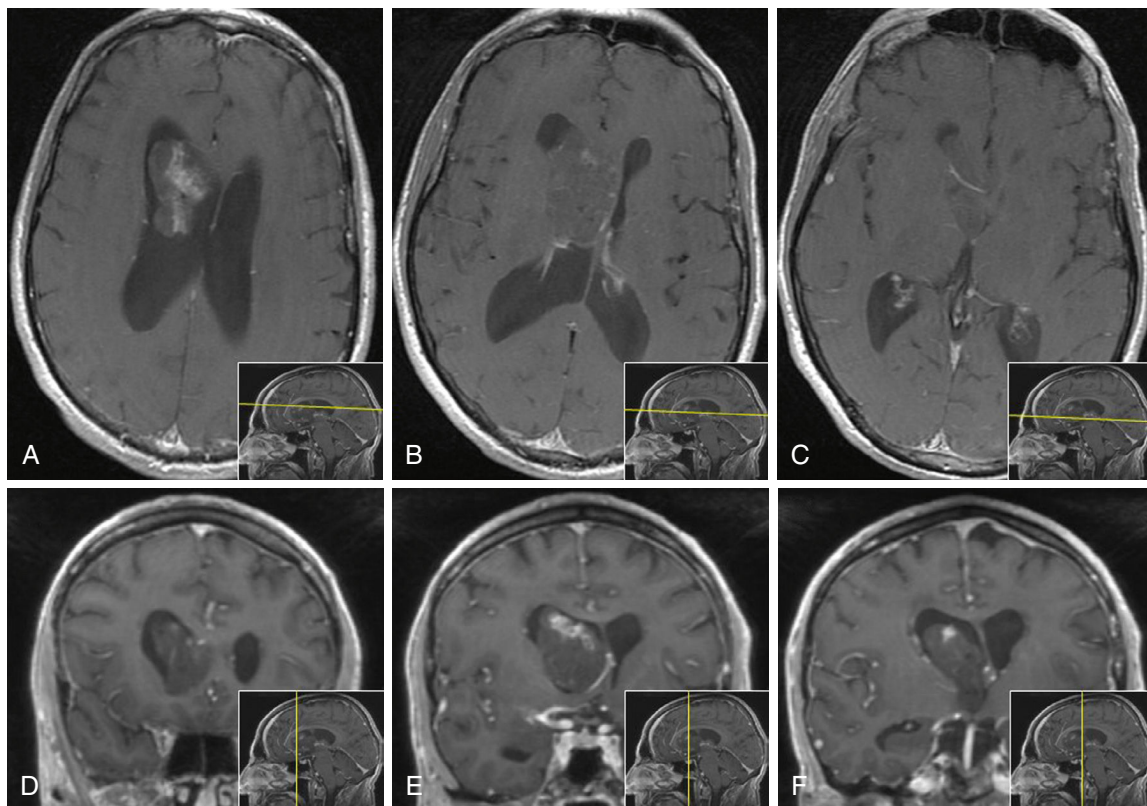


Figure 170.11. A 66-year-old male patient presented with headaches and altered mental status and was found to have a grade I subependymoma (A–F) following resection through a transfrontal transcortical approach to the right lateral ventricle. A foraminoplasty was performed to remove the tumor within the third ventricle (F).

Anterior Temporal Neocortical Resection

Limited resection of the anterior temporal lobe provides access to ventricular lesions located anterior to the level of the cerebral peduncle. Risks to the Meyer loop and the language cortices of the dominant temporal lobe are reduced compared with other variations of the anterior temporal approach. Unfortunately, the view following an anterior temporal neocortical resection is inherently limited in its posterior reach, and a bulky myocutaneous flap reflected anteriorly should be avoided to maximize anterior-posterior operative maneuverability.

Anterior Temporal Approach

The anterior transtemporal approach may be employed for tumors of the temporal horn or atrium, preferably of the nondominant hemisphere given the risk for receptive aphasia.^{27,46,48,55,60} One of the primary advantages of this approach is its short trajectory allowing ready access to the choroidal arterial pedicles, which often provide the vascular supply of temporal horn tumors, although special care should be taken to avoid sacrificing the anterior choroidal artery.^{11,61,62} Adjuncts including fMRI, DTI, and awake craniotomy may be used to avoid morbidity when planning this approach on the dominant side.⁶³ The position of the vein of Labbe should be considered on preoperative imaging and preserved during surgery.^{64,65} If a large corticectomy is made around the vein, we typically leave a bridge of nonfunctional cortical tissue beneath the vein to prevent its collapse and occlusion.

The patient is placed in the supine position with the head tilted laterally and the neck slightly extended. We generally prefer to use a linear or curvilinear incision. The temporalis is mobilized, and a craniotomy is performed down to the floor of the middle fossa. Using a combination of navigation and ultrasonography, the tumor is localized within the temporal horn. A corticectomy is subsequently made within the middle temporal gyrus, or a transsulcal approach between the middle and inferior temporal gyri is chosen. Although less common, some practitioners advocate for an inferior temporal gyrus corticectomy when working on the dominant hemisphere to decrease the risk for damaging the language cortex or inferior visual radiations that loop around the superior lateral aspect of the temporal horn.^{62,66–68}

Posterior Temporal Approach

The posterior temporal approach is best suited to access lesions in the posterior temporal horn and ventricular atrium. It aims to avoid damaging the periventricular optic radiations and cortical speech areas of the dominant temporal lobe through its inferior-to-superior-based trajectory.⁶⁹ Another major advantage is its near-immediate access to choroidal artery pedicles for tumor devascularization.^{7,27} Identification of the vein of Labbe's position and drainage location should be noted. A posterior location of the vein is common, however whether it is superior or inferiorly located is equally if not more important for this approach. Large atrial tumors that will require significant temporal lobe retraction place the vein at significant risk.

For this approach, the patient is positioned in the supine position with the vertex of the head down to allow for gravity retraction of the temporal lobe. A subtemporal craniotomy is performed to provide access to the posterior temporal lobe. The temporal lobe is elevated, and the dura is opened to expose the gyrus or sulcus selected for the approach. The location and approach should be confirmed using surgical adjuncts such as neuronavigation and intraoperative ultrasonography to guide dissection, exposure, and removal of the tumor. Upon closure, the mastoid air cells of the temporal bone should be obliterated and the dura closed watertight to prevent CSF leakage.⁷⁰

Supracerebellar Transtentorial Approach

The supracerebellar transtentorial approach to the ventricular atrium via the collateral sulcus provides access to the posterior medial lesions of the ventricular atrium. This approach requires a relatively long reach, and it may be difficult to treat large, firm lesions that significantly extend into the occipital horn, lateral ventricular atrium, or ventricular body. The advantages of this approach are that it spares fibers of the optic radiation, minimizes lateral neocortical injury, and may provide early access to choroidal feeding vessels for tumor devascularization. It is key that the long axis of the tumor lines up with the surgeon's view following entry through the collateral eminence. A tumor extending significantly out of this trajectory may be difficult to resect.

Transsylvian Approach

The transsylvian approach provides selective access to the temporal horn. The main benefits derived from this approach can be found in its ability to resect paraventricular lesions arising from mesial temporal lobe structures while sparing the lateral cortical surface gyri and visual fibers of the Meyer loop.^{19,71,72} Although anterior temporal exposure is adequate using this approach, it also provides limited access to the middle and posterior temporal horn after cortical entry is made below the inferior peri-insular sulcus.

Technical challenges typical of this approach revolve mainly around damage to the sylvian veins and vascular injury or vasospasm of the middle cerebral artery branches or their perforators. Excessive retraction on the operculum of the dominant frontal lobe during this approach may also result in expressive aphasia.

Occipital Transcortical Resection

The occipital transcortical approach to the occipital horn can provide the most direct access to the entirety of the occipital horn, especially if widely dilated because of trapping by the tumor. However, the risk for causing a contralateral visual deficit resulting from sectioning of visual fibers is very high. We selectively utilize this approach and mainly consider it for select patients with irreversible preoperative homonymous hemianopsia and significantly dilated ventricular horns. Furthermore, the vascular pedicles for tumors in this region often arise from the anterior medial aspect, which can make it difficult to access using this approach.

SURGERY OF THE THIRD VENTRICLE

Principles

Surgery of the third ventricle presents unique challenges to the operator because of a combination of the long reach required to access tumors in this area, limited operative working angles for handling tumors and achieving hemostasis, and myriad vital structures in this region. Some examples of such crucial structures include the thalami, hypothalami, fornices, infundibulum, optic apparatus, deep cerebral veins, and choroidal arteries.^{10,73}

Third ventricular tumors originate from three primary areas. Periventricular tumors arise from the sella, suprasellar, or pineal regions and extend into the third ventricle. Ventricular wall tumors arise from the neuroglia of the thalamus or hypothalamus. Intraventricular tumors arise from primary intraventricular structures (e.g., choroid plexus, telachoroidea) or ectopic tissue trapped within the ventricle.

The choice of approach to the third ventricle is based on trajectory and consists of four primary directions: anterior superior, subfrontal (including endonasal), lateral, and posterior.

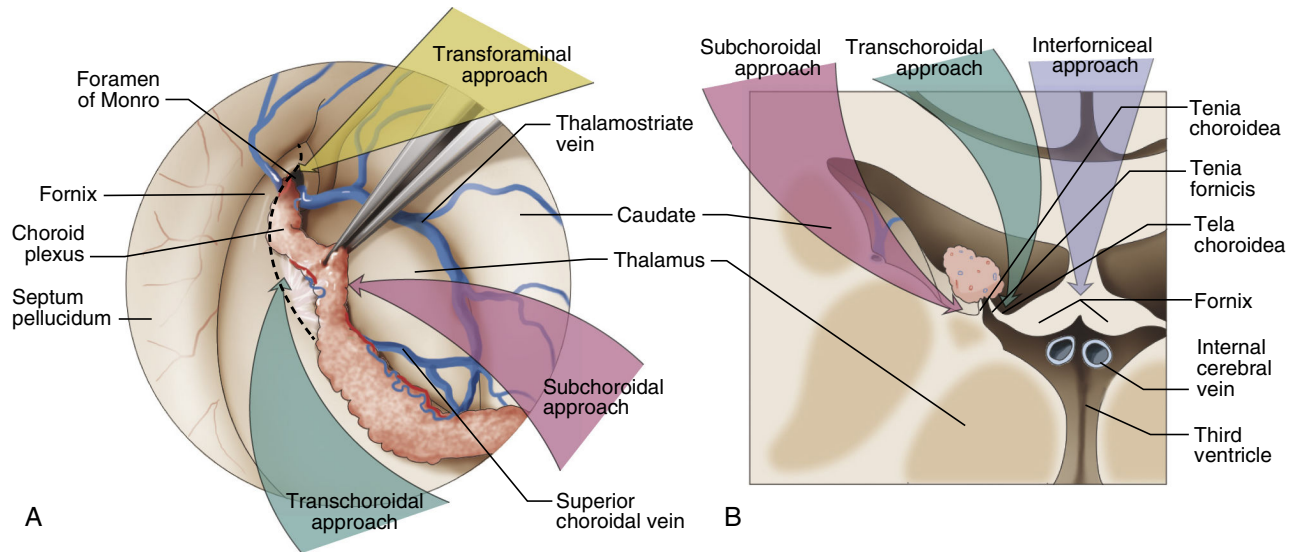


Figure 170.12. Techniques for safe entry into the third ventricle (A) from a superior approach following entry into the right lateral ventricle. Coronal view (B) showing the most frequently described approaches to the third ventricle: subchoroidal (red), transchoroidal (green), and interformiceal (blue) approaches. (Modified from Rhoton AL. *The lateral and third ventricles*. *Neurosurgery*. 2002;51[4 Suppl]:S207–S271.)

The anterior superior options are either midline (interhemispheric transcalsal) or paramedian (transcortical) and provide access via the lateral ventricles using a variety of techniques (Fig. 170.12). In general, we try to minimize tissue disruption by using the foramen of Monro as a natural orifice, expanding it posteriorly along the choroidal fissure if required.

The choice of surgical approach should integrate available knowledge including the goals of surgery, origin of the tumor, suspected pathology, and regional anatomy while remaining flexible to adapt to changes in treatment plan or unexpected complications (Table 170.3). A thorough understanding of third ventricular roof anatomy is required to safely execute the various techniques for entering the third ventricle from the lateral ventricles. A complete description of the anatomy of this region is provided in the “Surgical Anatomy” section earlier in this chapter. As a brief summary, the third ventricular roof is made up of five layers: the fornices, superior membrane of the tela choroidea, velum interpositum (a potential space containing the internal cerebral veins and medial posterior choroidal artery), inferior membrane of the tela choroidea, and third ventricular choroid plexus (see Fig. 170.5).⁷⁴

Equally important is knowledge of the major veins of the lateral ventricle and their relationship with the internal cerebral vein. The anterior septal, choroidal, and thalamostriate veins make up the primary venous drainage of the lateral ventricles, with the anterior septal and thalamostriate veins anastomosing with the internal cerebral vein in the region of the foramen of Monro, creating a “venous angle” on imaging, of which the exact location is variable.^{73,75} Generally, the anterior septal and choroidal veins may be sacrificed without consequence, however division of the thalamostriate vein has been reported to cause drowsiness, hemiplegia, mutism, basal ganglia infarction, and even death.⁷³ Although some report that this can be performed without ill effects, we prefer to spare this structure whenever possible.⁷⁶

A careful preoperative neuroendocrine and/or neuroophthalmological examination should be performed in cases of large suprasellar or anterior third ventricular tumors that exhibit compression of the optic apparatus, pituitary stalk, or hypothalamus. Endocrinopathies, especially of cortisol or thyroxine, should be corrected before surgical intervention. Because of the proximity of mesolimbic structures in this region, it

is recommended that a detailed preoperative neuropsychological assessment be performed if it can be completed expeditiously without delaying surgery.

Surgical Approaches

Transnasal Transsphenoidal Endoscopic Approach

The field of endoscopic transnasal surgery has experienced significant growth, with improved endoscope technology, specialized surgical instruments, and an extensive breadth of techniques allowing for increasingly difficult surgeries to take place with improved outcomes.⁷⁷ This approach is primarily chosen for tumors that originate in the sella or suprasellar region and extend into the third ventricle, such as craniopharyngiomas, Rathke cleft cysts, large tuberculum meningiomas, or giant pituitary macroadenomas (Fig. 170.13). One of the primary advantages of this approach is the ability to work along the long axis of the tumors in this region. Additionally, gravity may act as an aid in presenting the tumor into the surgical view for removal. Technical limitations of this approach include a narrow and long working corridor to the third ventricle that is restricted by the carotid arteries and optic apparatus. Extended skull base removal may help expand the reach of surgical instruments, however tumors that are firm, adherent, or encasing neurovascular structures still pose a particular challenge. A combined endonasal and transcranial approach may be considered, especially in cases with significant tumor extension lateral to the carotid siphon or if tumor is suspected to be particularly adherent to neurovascular or hypothalamic structures (Fig. 170.14). Difficulty achieving sharp dissection may require a greater degree of tumor manipulation that may lead to avulsion or vasospasm of superior hypophyseal branches to the optic nerve, resulting in ischemic injury and loss of vision, while dissection of tumor off of the hypothalamic tissue can cause profound endocrinologic, metabolic, or neuropsychiatric disruption.

Before embarking on the transnasal transsphenoidal endoscopic approach, extracranial CT and MRI are examined in regard to important features such as nasal turbinate anatomy, sphenoid sinus aeration, skull base erosion or hyperostosis, intercrotid distance, location of sphenoid septations, and the presence of an onodi cell containing the optic nerve. Intracranially, the surgeon

TABLE 170.3 Third Ventricle Approaches

Category		Regional Access				Advantages	Disadvantages	Risks
		Approach ^a	Anterior	Parasellar	Posterior	Pineal		
Anterior	Interhemispheric Transcallosal or Transfrontal	Transforaminal	☑				Utilizes natural opening through foramen of Monro Can sacrifice anterior septal vein to widen access to third ventricle Provides extensive access to the third ventricle	Limited working space through foramen of Monro Risk for short-term memory loss caused by fornix injury
		Transforaminal Transchoroidal	☑		☑			Attendant risks to deep venous drainage (especially internal cerebral veins) and fornix Attendant risks to deep venous drainage (especially thalamostriate, superior choroideal, and internal cerebral veins) and fornix
		Transforaminal Subchoroidal	☑		☑		Increased risk for injury to the thalamostriate and superior choroideal veins	Attendant risks to deep venous drainage (especially internal cerebral veins) and fornix Risks to neighboring neurovascular structures including optic apparatus, pituitary stalk, hypothalamus
		Interhemispheric Transcallosal	☑		☑		Provides extensive access to the third ventricle	Attendant risks to deep venous drainage (especially internal cerebral veins) and fornix Risks to neighboring neurovascular structures including optic apparatus, pituitary stalk, hypothalamus Risk for CSF leak, meningitis, or mucocoele if frontal sinus is violated
Lateral	Cranio-orbital	Subfrontal	☑	☑			Allows for access to the anterior-inferior aspect of the third ventricle, especially in the presence of a pre-fixed chiasm	Risks to neighboring neurovascular structures including optic apparatus, pituitary stalk, hypothalamus Risk for CSF leak, meningitis, or mucocoele if frontal sinus is violated
							Provides extensive access to the third ventricle	Risks to neighboring neurovascular structures including optic apparatus, pituitary stalk, hypothalamus Risk for CSF leak, meningitis, or mucocoele if frontal sinus is violated
Posterior	Occipital Transtentorial	Trans-Lamina Terminalis	☑	☑			Allows for greater access to suprasellar tumors with significant lateral extension than standard frontotemporal approach Allows for improved inferior-to-superior working angles compared with frontotemporal and subfrontal approaches	Limited view and working angle primarily to the anterior-inferior third ventricle Risks to neighboring neurovascular structures including optic apparatus, pituitary stalk, hypothalamus Risk for CSF leak, meningitis, or mucocoele if frontal sinus or temporal air cells are exposed and not reconstructed
					☑	☑	Provides access to pineal region and fourth ventricle if the deep veins are displaced inferoposteriorly Can access tumors spanning both supra and infratentorial compartments Versatile approach that can incorporate transfalxine and transplanial additions	Tumor removal may be challenging if deep veins displaced superiorly Manipulation of the medial occipital cortex Risk for venous injury to deep venous system (especially straight sinus, vein of Galen, internal cerebral arteries, basal veins of Rosenthal) Risk for visual compromise with occipital lobe manipulation

Category	Approach ^a	Regional Access			Advantages	Disadvantages	Risks
		Anterior	Parasellar	Posterior			
	Infratentorial Supracerebellar		☑	☑	Provides access to pineal region if the deep veins are displaced antero-superior No risk to visual cortices	Tumor removal may be challenging if deep veins are displaced inferoposterior	Risk for venous injury to deep venous system (especially vein of Galen, internal cerebral arteries, basal veins of Rosenthal)
	Interhemispheric		☑	☑	Avoids significant manipulation of the deep veins	Exposure limited primarily to the posterior third ventricle	Risk for venous injury to deep venous system (especially internal cerebral arteries, straight sinus, vein of Galen)
	Trans-Splenic		☑	☑	Superior access to tumors below the splenium adherent to the roof of the third ventricle or internal cerebral veins	Exposure limited primarily to the posterior third ventricle and sub-splenic region	Relatively high likelihood of causing disconnection syndrome compared with transcallosal approach
	Retro-Callosal		☑	☑	Versatile approach that can incorporate transfalcine and transsplenic additions to access tumors that traverse the supra and infratentorial compartments	Must work within and around the deep venous drainage system Manipulation of the medial occipital cortex	Risk for venous injury to deep venous system (especially straight sinus, vein of Galen, internal cerebral arteries, basal veins of Rosenthal)
Transnasal	Transnasal	☑	☑	☑	May provide the best working angle to long axis of the tumor Provides excellent access to sub-chiasmatic tumor; provides third ventricular access via the lamina terminalis	Tumors that extend lateral to the carotid arteries may be difficult to access and manage Tumors that are firm or stuck may not present themselves to the operator requiring a long reach through a narrow corridor Limited vision of structures that may be damaged outside of the direct view (e.g., vessel avulsion)	Risk for visual compromise with occipital lobe manipulation Endocrinologic dysfunction caused by damage to the pituitary gland, stalk, or hypothalamus CSF leak Damage to visual apparatus, optic nerve perforating vessels, or carotid arteries

^aBolded approaches represent commonly used "workhorse" approaches.

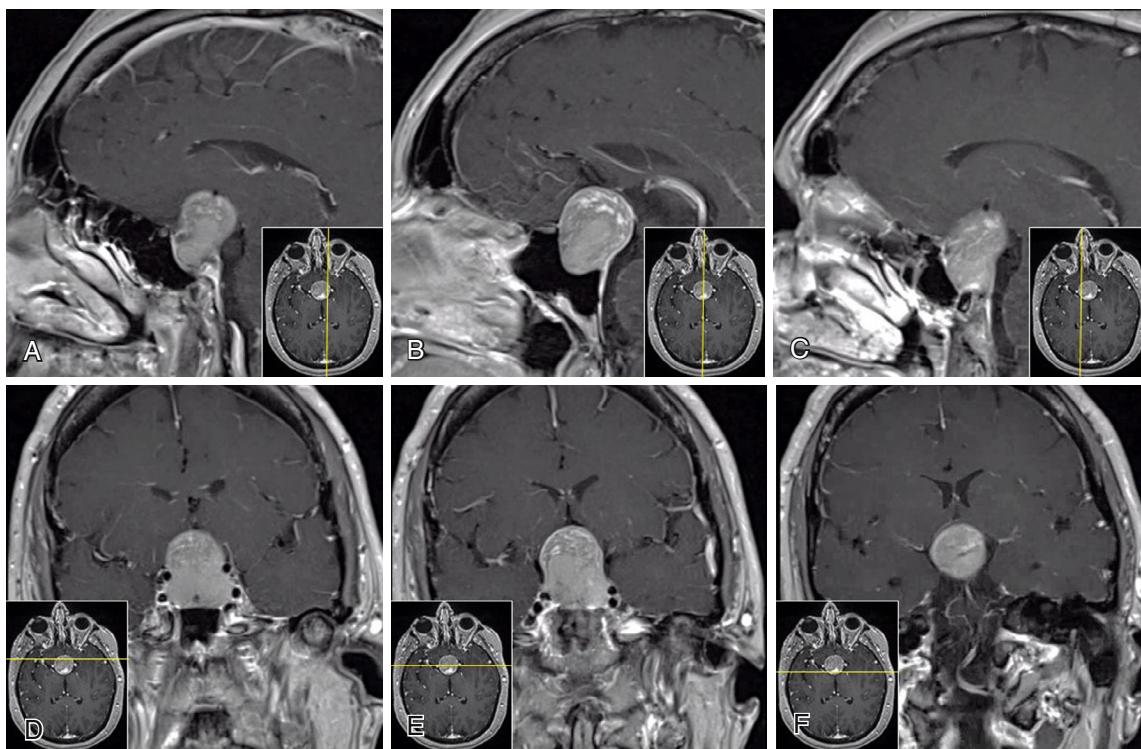


Figure 170.13. A 54-year-old male patient with worsening bitemporal visual loss was taken for an endonasal, transsphenoidal resection of a large prolactinoma (A–F) that was refractory to medical management.

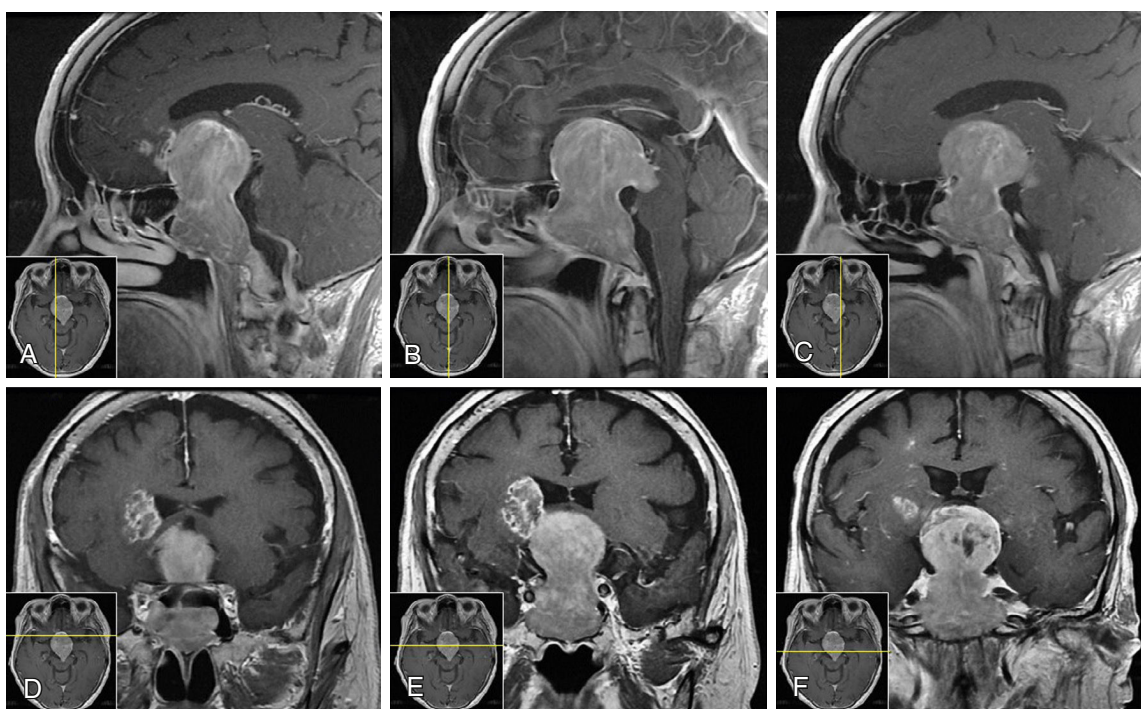


Figure 170.14. A 72-year-old female patient who presented with near-total blindness and hemodynamic instability secondary to panhypopituitarism elected surgical management for a macroadenoma (A–F). A combined endonasal and cranio-orbital approach was used for maximum safe resection of the tumor.

should study the size, angulation, and shape of the sella and tuberculum if considering a transnasal approach to decide on the extent of skull base removal required to approach the tumor. The location of the pituitary gland and stalk should be determined as best possible to avoid injury. Regardless of the anticipated

amount of bony removal, entry into the third ventricle results in a high-flow leak, and the surgeon should prepare to employ a vascularized flap for skull base reconstruction. The surgeon should also note the position of the optic chiasm preoperatively to anticipate whether the surgical view will be adequate following

extended skull base removal. This depends on the angulation of the chiasm, which may be influenced by the tumor; the location of the optic chiasm in the prefixed or postfixed locations (each occurs with a frequency of 10%); and the location of the pituitary gland and size of the sella (an enlarged sella with a posteriorly displaced pituitary gland may provide a favorable corridor for the transnasal transsphenoidal approach). If the chiasm is relatively low and postfixed with a favorable tuberculum angulation, approaching the lamina terminalis from between and above the optic nerves by removal of the anterior skull base may provide the best view of the third ventricle. On the other hand, if the optic chiasm is steeply angled and prefixed, then a subchiasmatic approach to the lamina terminalis and third ventricle will likely be favorable. Regardless, a large third ventricular tumor may “present” itself by stretching the lamina terminalis and providing a generous surgical window.⁷⁸

The patient is positioned supine with the head in extension and the chin slightly turned to the patient’s right so the dorsum nasi is parallel to both the floor and walls of the operating theatre; this allows the surgeon to maintain awareness of the patient’s anatomy and the operative instruments. The foot of the bed is angled and positioned slightly toward the left for a right-handed surgeon to be able to stand and operate comfortably. A rescue nasal septal flap is harvested before sphenoidotomy. The sphenoid sinus is opened widely, and vital structures are identified. Frequently, the intrasphenoidal septations will intersect with either of the two carotid arteries. We use micro-Doppler imaging combined with neuronavigation to confirm the location of the cavernous carotid artery. The subsequent steps vary significantly based on the individual tumor. Following removal of sphenoid mucosa and posterior ethmoidectomies to allow for unfettered use of the endoscopic surgical instruments, the bone of the posterior planum, tuberculum sellae, and upper portion of the sella is removed. If possible, the sella floor is preserved to allow for skull base reconstruction. Copious irrigation is used to prevent thermal injury to the optic nerves or their feeding vessels during tuberculum drilling. The dura is opened in a cruciate fashion, and bleeding from the superior intercavernous sinus is controlled. Transposition of the pituitary gland may be performed if necessary. After opening the dura and diaphragm and accessing the suprasellar cistern, tumor debulking and removal may commence. Microsurgical technique and sharp dissection should be used when safe to avoid excess traction on cranial nerves. In the event of a large amount of tumor remaining difficult to reach or hidden from surgical view, a controlled Valsalva maneuver may be used to force the tumor into the operative field. We recommend the use of an angled endoscope to confirm tumor removal before closure.

After the goals of surgery have been achieved, closure may commence. It is imperative that a watertight closure is performed to prevent CSF leakage. Depending on the size of the defect, a combination of grafting materials or a vascularized flap may be used. Despite improvements in technology and closure techniques, postoperative leak rates are estimated to occur in 10% to 20% of cases.^{79–82} Following surgery, patients should continue to be closely monitored for the development of endocrinopathies, a preventable cause of postoperative morbidity and mortality.

Subfrontal Approach

The lamina terminalis makes up the anterior wall of the third ventricle and consists of a thin membrane composed of gray matter, pia mater, and ependyma that extends from the anterior commissure to the superior aspect of the optic chiasm. The subfrontal translamina terminalis approach provides relatively confined access to the anterior third ventricle, defined by a triangular space anterior to a line drawn from the foramina of Monro to the sylvian aqueduct (Video 170.3).⁸³ It is primarily used for smaller

lesions that are located primarily in the third ventricle with only a moderate suprasellar component because it provides limited subchiasmatic access. Ideally, the third ventricular tumor will enlarge the lamina terminalis as operating through a nonexpanded lamina terminalis is highly restrictive and places the patient at risk for endocrinologic and cognitive deficits as a result of manipulation of the hypothalamus.

The patient is placed in the supine position with slight head extension to allow for gravity retraction of the frontal lobe with CSF release. Multiple variations on the subfrontal approach exist utilizing different types of craniotomies. The primary methods used are the bifrontal and unilateral frontal craniotomies. Each method provides a different angle of approach as well as varying degrees of gravity retraction following arachnoid release. Next, an incision tailored to the chosen approach is made. This is typically a bicoronal, frontotemporal, or eyebrow incision. Here, we will share our general technique for the interhemispheric subfrontal translamina terminalis approach (Video 170.4).

A bicoronal (Souttar) incision is made, and pericranium is preserved to allow for reconstruction. If a unilateral frontal craniotomy is chosen instead, navigation should be used to help localize and avoid the frontal sinus as the supraorbital notch does not provide a highly reliable method of delineating the lateral extent of the frontal sinus.⁸⁴ A bifrontal craniotomy is performed, with significant attention on getting as low as possible along the floor of the anterior fossa to prevent the need for further drilling. The frontal sinuses are exenterated using the surgeon’s method of choice. The roof of the orbit is drilled flat to maximize the operative working space of the surgeon. The dura is opened parallel to the orbital roof on either side of the anterior limb of the SSS. The sinus is ligated, and the insertion of the falx is resected from the crista galli. The olfactory bulbs and tracts are sharply dissected from the olfactory sulcus at the base of the frontal lobes to minimize traction injury leading to anosmia. Following the olfactory tracts back toward the suprasellar region, the ipsilateral anterior clinoid and optic nerve are encountered. Opening of the basal cisterns at this location allows for CSF egress, with wide arachnoid dissection permitting further retraction of the frontal lobes. With adequate exposure, the operator can identify the optic nerves, optic chiasm, pituitary infundibulum, anterior communicating artery, and the paired anterior cerebral, middle cerebral, posterior communicating, and internal carotid arteries. The lamina terminalis, which normally has a “reddish blush,” may be opened to the genu of the corpus collosum using sharp dissection. Care must be taken to avoid injury to the optic apparatus and its blood supply.³⁸ Sharp dissection is preferred for separating the tumor from the nerve, and bipolar cautery is prohibited. When the tumor is adjacent or invading the lamina terminalis, it often adapts a gray discoloration, which helps distinguish it from the surrounding optic nerves. Tumor debulking and removal through the lamina terminalis can now take place. If there is tumor in the subchiasmatic region, it can be removed by operating around the optic chiasm through areas such as the prechiasmatic space or optocrotid triangle.^{38,85}

Transfrontal Approach

The transfrontal approach is a workhorse approach to lesions of the third ventricle, especially when facilitated by dilated lateral ventricles from obstructive hydrocephalus. The initial steps of this approach to access the lateral ventricle was described previously. When coupled with the transforaminal or transchoroidal technique, the majority of the third ventricle can be reached using this approach as can lesions extending laterally into the wall of the third ventricle. The “blind spots” of this approach are primarily anterior-ventral near the chiasmatic recess in the suprasellar region and posterior-superior in the pineal region. The ipsilateral ventricle is entered via an oblique trajectory from the



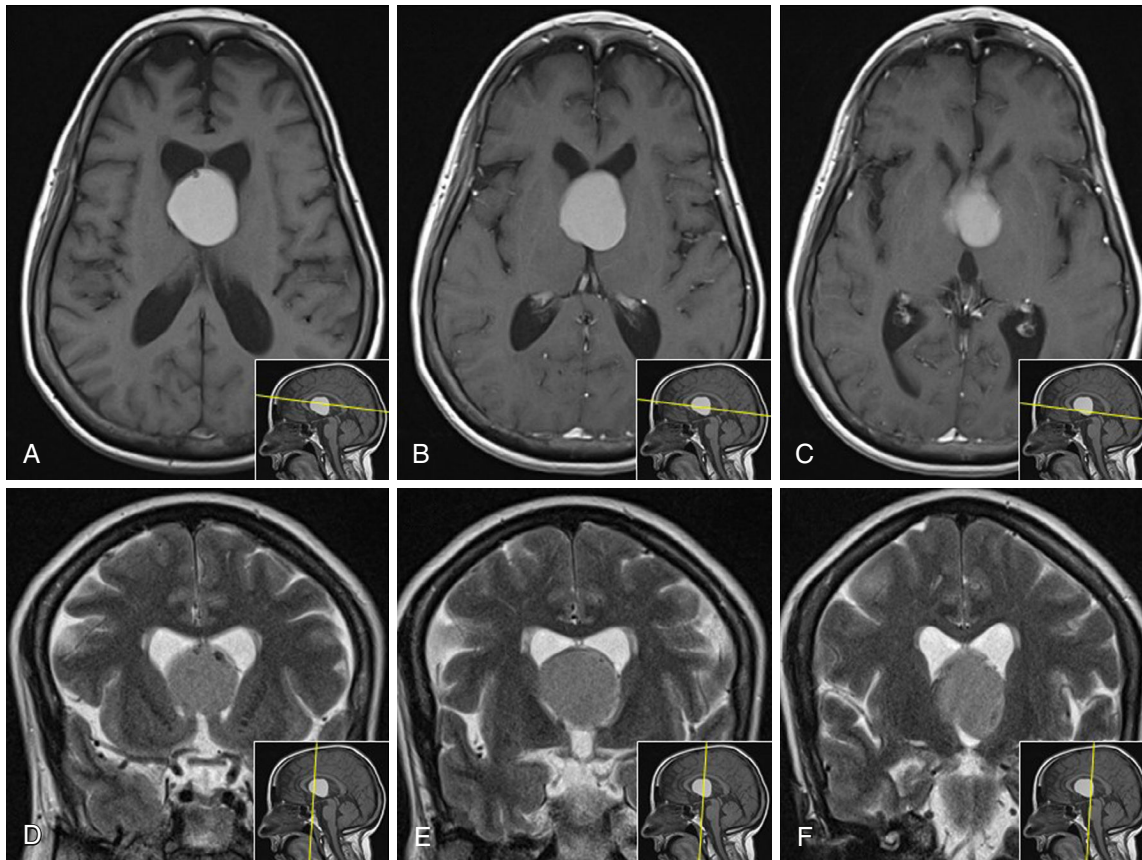


Figure 170.15. A 56-year-old female patient with recurrent episodes of intractable nausea and vomiting underwent a right transfrontal transforaminal approach with foraminoplasty for resection of a third ventricle colloid cyst (A–F).

lateral cortical surface. Upon entry into the lateral ventricle, standard methods of reaching tumors of the third ventricle may be pursued, such as the transforaminal and transchoroidal methods (Fig. 170.15). We favor the use of the transfrontal approach for a variety of third ventricular tumors, especially those that originate along the roof (e.g., colloid cyst, meningioma) and expand the foramen of Monro. The slightly lateral approach allows the operator to see across the ipsilateral foramen to visualize the third ventricular roof and the contralateral foramen of Monro. We do not typically couple this approach with the interforaminal technique as we believe it increases the risk for fornix injury given the lateral-to-medial operative trajectory.

Anterior Interhemispheric Transcallosal Approach

The anterior interhemispheric transcallosal approach to the lateral ventricle was described previously in the discussion of lateral ventricular tumors. As with the transfrontal approach, there are numerous methods of accessing the third ventricle using the interhemispheric approach (Fig. 170.16). The most frequently used techniques include the transforaminal, interforaminal, subchoroidal, and transchoroidal approaches (Fig. 170.17 and Video 170.5). Regardless of the technique selected, the fornix and vessels traveling within the velum interpositum are at risk for injury, especially the internal cerebral and thalamostriate veins and the medial posterior choroidal arteries. There is a blind spot at the anterior inferior aspect of the third ventricle that can be accessed through the foramen of Monro if adequately enlarged by a giant tumor, or through the transforaminal approach. However, given the risk for damaging the fornices and causing memory loss, we rarely consider the extra anteroventral exposure afforded by this route to be

worth the risk.^{73,86–89} This area of the third ventricle can be safely reached through a variety of other approaches described in this chapter.

Following a standard anterior interhemispheric transcallosal approach as described previously, the operator will enter the lateral ventricle. After confirming orientation by using the same lateral ventricular anatomic structures as mentioned previously, the surgeon will evaluate any visible tumor and regional anatomic distortion to decide whether the preoperative plan should be altered. Irrespective of which of the following techniques are selected, the velum interpositum, a potential space located between the two thin membranous layers of the tela choroidea containing the internal cerebral veins and medial posterior choroidal arteries, must be transgressed to reach the third ventricle. The internal cerebral veins must be identified early and carefully preserved during dissection and throughout the entirety of the operation.

Transforaminal Technique. The transforaminal approach takes advantage of a widened foramen of Monro to allow for tumor debulking and removal through this anatomic route without the need for expanded access. However, the fornix nearly completely encircles the foramen, and care must be taken to minimize both traction and direct injury to this structure. Even unilateral injury has been shown to cause anterograde amnesia.⁹⁰ This technique is commonly used for tumors of the third ventricular roof and for those that are amenable to debulking or early fenestration to allow for easier tumor removal through this natural orifice. If wider exposure is needed, a foraminoplasty may be performed. This can be achieved in two ways: through the sacrifice of the anterior septal vein with enlargement of the foramen or via

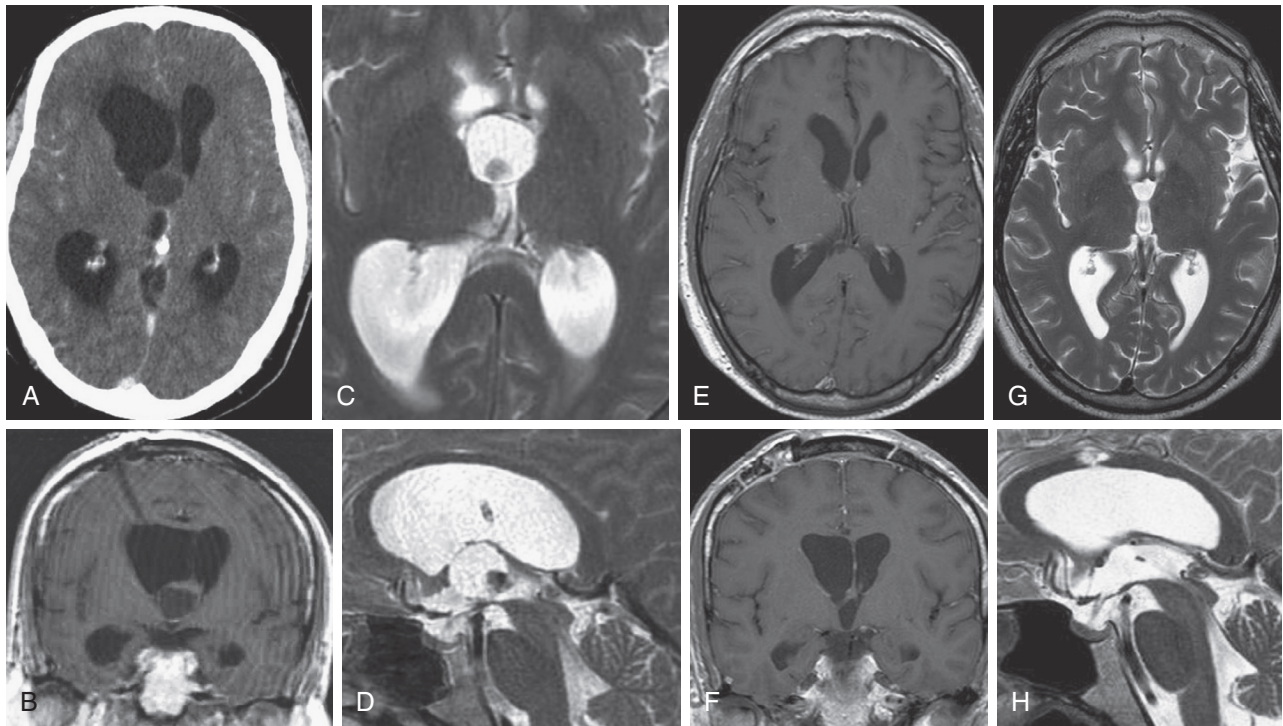


Figure 170.16. A 51-year-old male patient was admitted on an emergency basis for obstructive hydrocephalus, as seen on preoperative CT scans (A–B). Initially, an external ventricular drain was placed to release the raised intraventricular pressure. Preoperative T2-weighted MR images in the axial plane (C) and sagittal plane (D) show a large colloid cyst within the third ventricle obstructing the foramen of Monro. Complete removal of the colloid cyst was achieved microsurgically via the right frontal interhemispheric transcalsal approach and confirmed on postoperative T1-weighted MR images (E–F) and T2-weighted MR images (G–H). The patient's postoperative course was entirely uneventful.

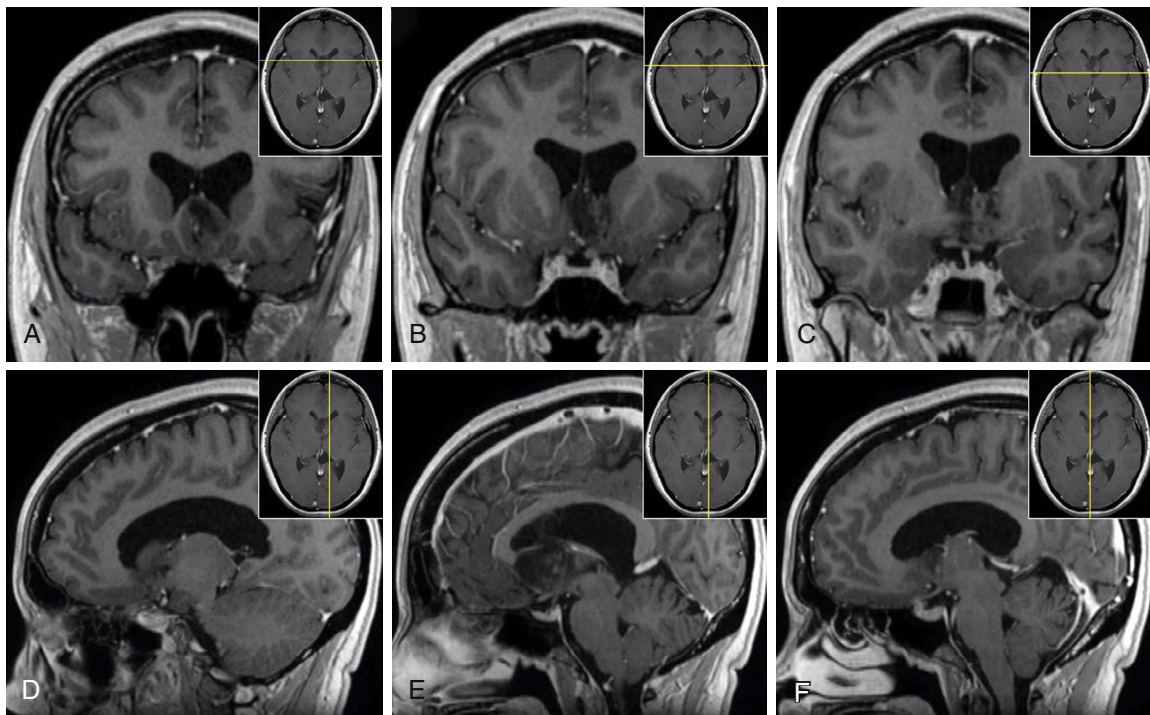


Figure 170.17. A 65-year-old female patient with a history of metastatic breast carcinoma underwent an interhemispheric transchoroidal approach for resection of a metastatic periventricular lesion (A–F) extruding into the third ventricle.

the transchoroidal approach to the third ventricle. Of note, the classic description of the junction between the anterior septal, thalamostriate, and internal cerebral veins is that it lies immediately adjacent to the posterior margin of the foramen of Monro; however, Türe et al. described its actual location to be variable, occurring further posterior to the foramen in a significant number of cases, allowing for safe foraminoplasty without venous sacrifice. As the junction between the anterior septal vein and the internal cerebral artery can be delineated on preoperative venography, they advocated for precise planning on which side should be used when considering the transforaminal approach with foraminoplasty.⁷³

Transchoroidal Technique. As mentioned earlier, the transforaminal technique may be coupled with a transchoroidal approach to the third ventricle. The opening can extend from the foramen of Monro to the atrium of the lateral ventricle or anywhere in between. Described by Wen et al. in 1998, this approach utilizes the choroidal fissure as a natural route to enter the third ventricle, while mitigating risk to surrounding neurovascular structures.⁷⁴ As previously mentioned in the “Surgical Anatomy” section earlier in this chapter, the choroid plexus exists in continuum between the roof of the third ventricle and the floor of the lateral ventricle as it passes through the foramen of Monro and courses posteriorly through the ventricular body and then laterally into the temporal horn. Within the body of the lateral ventricle, the choroid plexus runs within the choroidal fissure, a natural cleft between the thalamus and fornix.⁹¹ The choroid plexus is attached to the ventricular floor by two layers of continuous ependyma called *taeniae* (meaning “band” in Latin) within the ventricular cavity. Medially it is attached to the body of the fornix via the taenia fornices, while laterally it attaches to the thalamus via the taenia choroidea. We prefer to release medial to the choroid plexus via the taenia fornices to access the tela choroidea.

The next key step requires careful dissection between the internal cerebral veins traveling between the layers of tela choroidea. There are two key reasons for splitting the internal cerebral veins. First, it avoids the thalamostriate and choroidal veins that run medial from the thalamus toward the internal cerebral veins draining the internal capsule and medial hemisphere. Second, any bridging veins between the two internal cerebral veins may be safely ligated without consequence. When choosing to utilize fixed retractors, the operator must take special care to avoid compressing the internal cerebral veins.

Finally, dissection continues through the last two layers of the roof of the third ventricle through the second layer of tela choroidea and then the choroid plexus that attaches to the roof of the third ventricle. By this time, the roof of the third ventricle has been completely transgressed, and the incision can be widened from the posterior aspect of the foramen of Monro to the area immediately anterior to the pineal gland.

Subchoroidal Technique. The subchoroidal approach to the third ventricle is similar to the transchoroidal route, except that the more lateral taenia choroidea is opened to access the velum interpositum. As a result, veins draining to the internal cerebral vein, such as the thalamostriate vein and occasionally the choroidal veins, as well as arteries, such as the medial posterior choroidal artery, must be navigated or sacrificed to obtain an intravenous route to the third ventricle.⁵⁹ Although one is further from the fornix, potentially decreasing the risk for damaging this structure, there exists a theoretical increased risk for vascular injury. Therefore, we favor approaching these tumors by releasing the taenia fornices medial to the choroid plexus using a transchoroidal approach. In addition, an anatomic study by Bozkurt et al. in 2018 described a more limited exposure provided by the subchoroidal technique when compared with the transchoroidal dissection.⁹²

Interforniceal Approach

The interforniceal approach to the third ventricle, based on techniques developed by Dandy and van Wagenen, was first described by Busch in 1944 because of frustration with the limited exposure provided by the traditional transforaminal route.⁹³ His technique was initially trialed with great success on a giant colloid cyst causing upward displacement of the fornices and thinning of the median raphe. This approach provides wide exposure of the third ventricle but runs the risk for injuring both forniceal bodies.^{94,95} Taking a midline interhemispheric transcalsal approach rather than a lateral transcortical approach provides a less biased view and working angle to both sides of the ventricle. This helps minimize forniceal retraction that often results from the laterally biased cortical entry point, thus reducing the risk for forniceal injury.⁹⁶ Doing so reduces the risk for cognitive impairment and memory loss.

Ideally, the presence of a septum cavum pellucidum provides a widened corridor for interforniceal dissection following an interhemispheric approach. If entry occurs into either ventricle, lateral to the septum, correct orientation should first be established via the usual landmarks, followed by identification of the forniceal body and midline septum. If the leaves of the septum are not fused, then the potential space between them can be entered and followed down to the midline forniceal raphe. However, if they are fused, then the septum must be fenestrated and subsequently coagulated to reduce it to the level of the midline raphe using bipolar coagulation. The point of intersection between these two structures (the septum pellucidum and fornix) serves as reliable guidance to the midline between the fornices. Special care must be placed on this relationship as the midline may be distorted in the presence of large third ventricular tumors. Starting between the two foramina of Monro, the raphe is then expanded posteriorly using sharp dissection and bipolar cautery for 1 to 1.5 cm. Further dissection posteriorly risks damaging the forniceal commissure.^{57,65} The remaining layers of the diencephalic roof are dissected to access the tumor. Often, large tumors will splay the critical vessels of the velum interpositum to either side, offering a safe corridor for dissection. However, preoperative imaging should be carefully studied to ensure that the internal cerebral veins and medial posterior choroidal arteries are not displaced to a single side, which greatly alters the chosen approach.

Posterior Interhemispheric Transcallosal Approach

The posterior interhemispheric transcallosal approach to the third ventricle provides access to most of the third ventricle, however it is limited in its ability to tackle lesions that have significant extension posteriorly or that heavily involve the pineal gland as it is in the blind spot of the operator's view and working angle. If necessary, a short segment of the splenium may be divided to extend the operative view posteriorly, but this risks a disconnection syndrome.

The posterior transcallosal approach follows the tenets of posterior midline interhemispheric dissection described previously. Following careful microsurgical dissection of the interhemispheric falx, the corpus callosum is reached and divided 2 to 3 cm anterior to the tip of the splenium. Occasionally, thinning of the corpus callosum and underlying hippocampal commissure may result from mass effect from the tumor or hydrocephalus, allowing for increased ease of dissection. Following the posterior callosotomy, the tela choroidea is encountered along with the internal cerebral veins. These are spared at all costs, and careful dissection of the surrounding arachnoid and tumor capsule is performed to allow for their mobilization if necessary. As mentioned previously, some authors believe that a limited division of the splenium can safely occur to allow for increased posterior access. In general, we prefer to

use an angled endoscope and instruments or a employ a different surgical approach when access to the posterior third ventricle and pineal region is necessary. Following tumor removal, the cavity is carefully inspected for residual using the angled endoscope, and the sylvian aqueduct is visualized to ensure patency.

Posterior Interhemispheric Retrocallosal/Occipital Transtentorial Approach

The retrocallosal approach to the posterior third ventricle and pineal region, described by Jamieson in 1971,⁹⁷ is often combined with a transtentorial incision to allow for additional access to the quadrigeminal cistern, superior cerebellar vermis, and deep galenic venous system. Displacement of the deep veins posterior-inferiorly by a large tumor provides a favorable pathway for the retrocallosal route by allowing the operator to readily work above and between the deep veins to remove posterior third ventricular and pineal region tumors (Video 170.6).⁹⁸ Although this approach offers a relatively wide surgical corridor given the deep location of these tumors, serious complications may occur with technical mishaps. Reports of mutism, flat affect, coma, and death have been reported in relation to injury to the internal cerebral veins, vein of Galen, or straight sinus.⁹⁹ Retraction or injury to the cortex of the medial occipital lobes can cause contralateral homonymous hemianopsia.¹⁰⁰ In addition, if the splenium is injured along with the dominant visual lobe, the patient may develop alexia without agraphia (i.e., pure alexia), which generally favors a right-sided approach presuming it is anatomically and ergonomically favorable for tumor resection. We avoid the use of fixed retractor blades given their risk for damaging the cuneus or lingual gyrus.

The patient is positioned in the lateral park bench position to allow for gravity retraction of the ipsilateral occipital lobe from the falx. The neck is placed in slight flexion with the SSS parallel to the ground. Before draping, the midline SSS and ipsilateral transverse sinus are marked and confirmed using navigation. The incision depends on the surgeon's choice, but we typically prefer a linear incision given its superior healing characteristics, especially if adjuvant radiotherapy will be required. An occipital posterior interhemispheric craniotomy is performed, fully exposing the midline SSS as well as the torcula and ipsilateral transverse sinus. Anatomically, there is a paucity of veins in this posterior occipital region, making this approach consistent and safe.¹⁰¹ However this does not obviate the need for careful review of venous anatomy before surgery, including determination of the laterality of the dominant transverse sinus. The dura is opened in a C-shaped fashion, hinging along the SSS. A stitch may be placed through the superficial falx to allow for gentle retraction of the sinus dura. Overzealous retraction should be avoided to avoid thrombosing the SSS. Interhemispheric dissection proceeds, and the straight sinus is identified. Of note, although it is often depicted as a midline structure, the straight sinus often takes an alignment that is slightly lateral to the true midline. An incision is made about 1 cm lateral along the tentorium and parallel to the straight sinus until connected with the edge of the incisura. Injury to intradural veins or entry into a venous lake may lead to brisk bleeding, which is best controlled by bipolar cautery or surgical clips. Although some advocate for ligation of the inferior sagittal sinus to allow for further transection of the falx, we do not regularly perform this maneuver unless an expanded falcine window becomes truly necessary. Sutures are used to reflect the free edges of the cut tentorium to maximize the operative view. At this time, the superior cerebellar surface as well as a thick arachnoid layer covering the deep cerebral veins are identified. This thickened arachnoid layer is opened using sharp dissection, taking care to avoid tearing that could injure deep venous structures. Systematic identification of the vein of Galen, internal cerebral veins, and basal veins of Rosenthal may

now be performed. Anterior venous displacement by the tumor makes early venous identification difficult and increases the risk for vascular injury. Following identification of the veins, the arachnoid layer can be further dissected to allow for mobilization of these veins for a widened operative corridor. At this point, tumor biopsy may be performed with frozen section analysis to dictate how aggressive to regard tumor resection. Once the goals of surgery are achieved, closure may commence with a goal for watertight closure of the dural flaps.

Infratentorial Supracerebellar Approach

The infratentorial supracerebellar approach to the third ventricle is a workhorse approach to the pineal and tectal regions as well as the posterior third ventricle. First described by Krause in 1926,¹⁰² this approach is most suitable for pineal region tumors or third ventricular lesions that expand the posterior wall, allowing the tumor to come to the surface at the level of the quadrigeminal cistern (Fig. 170.18 and Video 170.7). A favorable operative corridor is obtained ventral to the deep veins when they are displaced superior and anterior, allowing for a direct pathway by which to resect the tumor. When deciding whether to use a supratentorial or infratentorial approach, we draw a line from the pineal gland to the torcula. If the long axis of the tumor is below this line, then we choose to utilize the supracerebellar infratentorial approach. Otherwise, a supratentorial approach is favored. Although the supracerebellar approach spares the corpus callosum, parietal, and occipital lobes, complications related to cerebellar injury such as ataxia or cerebellar mutism syndrome have been reported.¹⁰³ Manipulation of the tectum may also induce ocular disturbances, however, these are often reported to be transient.¹⁰⁴ Recently, there has been increased usage of the paramedian supracerebellar infratentorial approach. Although it does not utilize the midline trajectory for anatomic orientation, proponents tout its shorter route and avoidance of the steep trajectory introduced by the apex of the culmen. In addition, it allows for sparing of midline vermian veins, and there is a decreased risk to the dominant transverse sinus by taking a contralateral approach beneath the lesser transverse sinus.^{105,106} Relative contraindications to this approach include significant dorsal or lateral tumor extension in a firm tumor that cannot be accessed despite a tentorial window, or displacement of the deep venous structures in the posterior-inferior direction obscuring safe access to the tumor.

When considering positioning for surgery, the angle of the tentorium and ability to flex the patient's neck should be taken into account. A steeper angle to the tentorium generally requires more flexion. Although either the sitting or Concorde position can be used, we prefer to use the Concorde position to maximize operator ergonomics. A midline linear incision is made, with careful dissection following the avascular nuchal ligament that exists between the trapezius, semispinalis capitis, and rectus capitis posterior minor. Dissection down to the midline posterior tubercle of C1 is not typically required. The extent of the craniotomy varies based on the pathology. A wider craniotomy allows for increased arachnoid dissection along the lateral recesses, which allows the cerebellum to fall away more readily to increase working space. In addition, a wide corridor allows for instruments to more easily assist each other and provide lateral traction when required. The craniotomy should be taken far enough superiorly to reveal the bilateral transverse sinuses and midline torcula. These structures are unroofed to allow for gentle manipulation if needed. The craniotomy is taken far enough inferior that CSF from the cisterna magna can be released before dural opening to allow for cerebellar relaxation, which facilitates upcoming arachnoid dissection. We open the dura in a Y-shaped fashion, reflecting the superior triangle upward to allow for an adequate view of the operative corridor. Supracerebellar dissection proceeds with removal

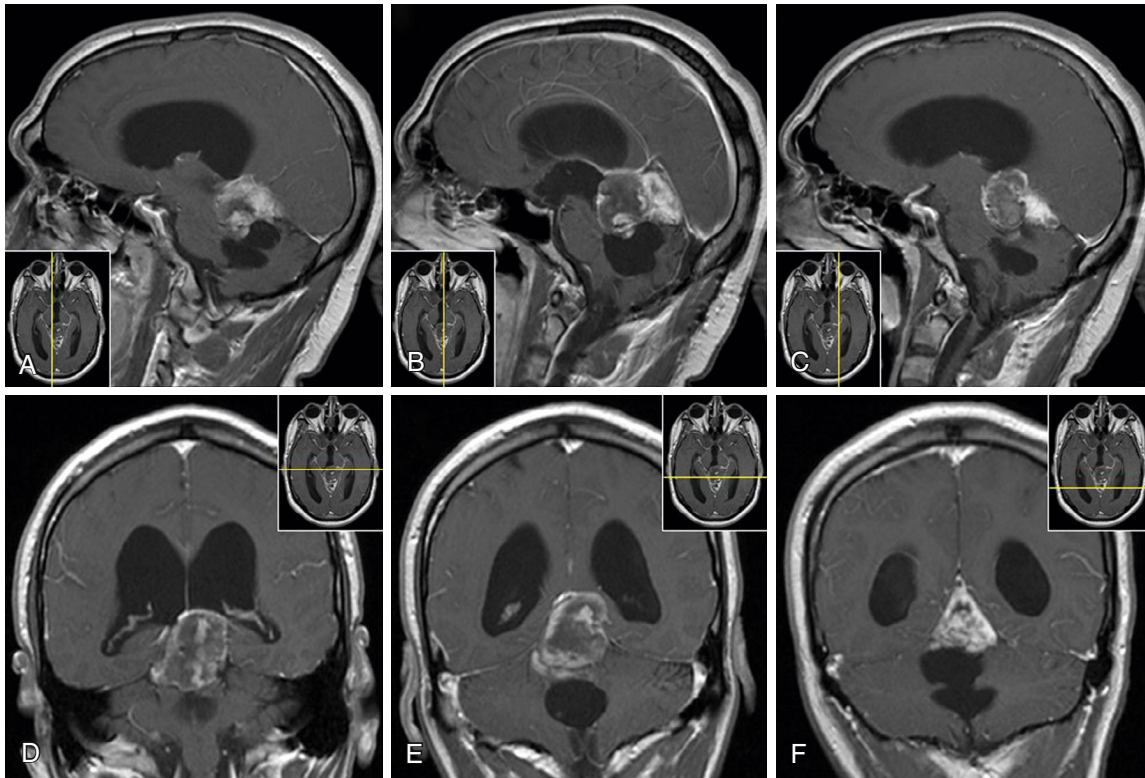


Figure 170.18. A 37-year-old male patient presenting with 2 months of blurry vision underwent a right frontal endoscopic third ventriculostomy, followed by a supracerebellar infratentorial approach for tumor (A–F) removal.

of intervening arachnoid bands and ligation of bridging veins. Releasing arachnoid bands at the lateral corners will allow for increased cerebellar retraction and a larger operative field. At no point do we use fixed retractors on either the tentorial or cerebellar surfaces as they can cause venous occlusion and risk damage to the cerebellar cortex and underlying tissue. Once an adequate degree of the superior cerebellar surface is exposed, a wet collagen sponge or patty is placed over its surface to protect it from the insertion and removal of surgical instruments and the heat from the microscope light. After crossing the apex of the cerebellar surface, a thick layer of arachnoid can be seen, beneath which lies the deep draining veins. This is the same arachnoid that is encountered from the retrosigmoidal approach but is seen from a ventral view. Often, the precentral cerebellar vein, which runs from the superior vermis to the vein of Galen, can be identified through the arachnoid layer. Given its inconvenient location in the center of the operative field, it may be coagulated and removed without consequence. Using sharp dissection of the arachnoid, the deep veins are exposed. This arachnoid covering may act to suspend the cerebellum, therefore its wide removal aids in cerebellar relaxation. The vein of Galen and lateral draining veins of Rosenthal are usually first to be identified with upward displacement of the internal cerebral veins. Small arterial branches of the medial posterior choroidal and superior cerebellar arteries may be noted as well. We spare these feeding arteries unless their ligation is absolutely necessary because of direct tumor involvement. By this time, the posterior face of a third ventricular or pineal region tumor should be visible, and tumor resection may proceed. Care must be taken not to injure the oft hidden-from-view internal cerebral veins that run superior to the tumor or are encased within it. Angled endoscopes are used to confirm the extent of resection followed by immaculate hemostasis. Dural closure must be completed in a watertight fashion to avoid pseudomeningocele formation and CSF leak.

Cranio-orbital Approach

The cranio-orbital approach to the third ventricle provides a suprachiasmatic view similar to the subfrontal approach. However, given its additional lateral trajectory, a more direct view of a tumor in the subchiasmatic region can be obtained through the opticocarotid triangle. Removal or thinning of the sphenoid bone (i.e., orbital roof, lateral orbit, lesser wing of the sphenoid) allows for a greater inferior-to-superior working angle within the third ventricle through the lamina terminalis. This makes it slightly more versatile for accessing large suprasellar tumors with significant third ventricular extension than a standard subfrontal or bone-sparing frontotemporal approach. When performing this approach, early access to the basal cisterns for early CSF release is key for obtaining the necessary retraction. Wide release of arachnoid bands or other adhesions also facilitates a widened operative corridor (Fig. 170.19).

An expansion on the cranio-orbital craniotomy, the cranio-orbitozygomatic approach takes advantage of an additional key osteotomy of the zygomatic process of the temporal bone. This additional bone removal allows for improved access along a subfrontal trajectory by allowing for inferior retraction of soft tissue structures. Risks associated with these frontolateral approaches relate primarily to the regional anatomy, including visual deficit, hypothalamic injury, hemiplegia, and endocrinologic disturbances. Furthermore, given the extent of bone removal, especially with the cranio-orbitozygomatic approach, CSF leak or enophthalmos may occur if proper skull base reconstruction is not performed. Damage to the frontal branches of the facial nerve and temporomandibular joint can be avoided with proper understanding of local anatomy.

The technical description of how to perform the cranio-orbital and cranio-orbitozygomatic approaches is described elsewhere. Following bone-flap removal and dural opening, the

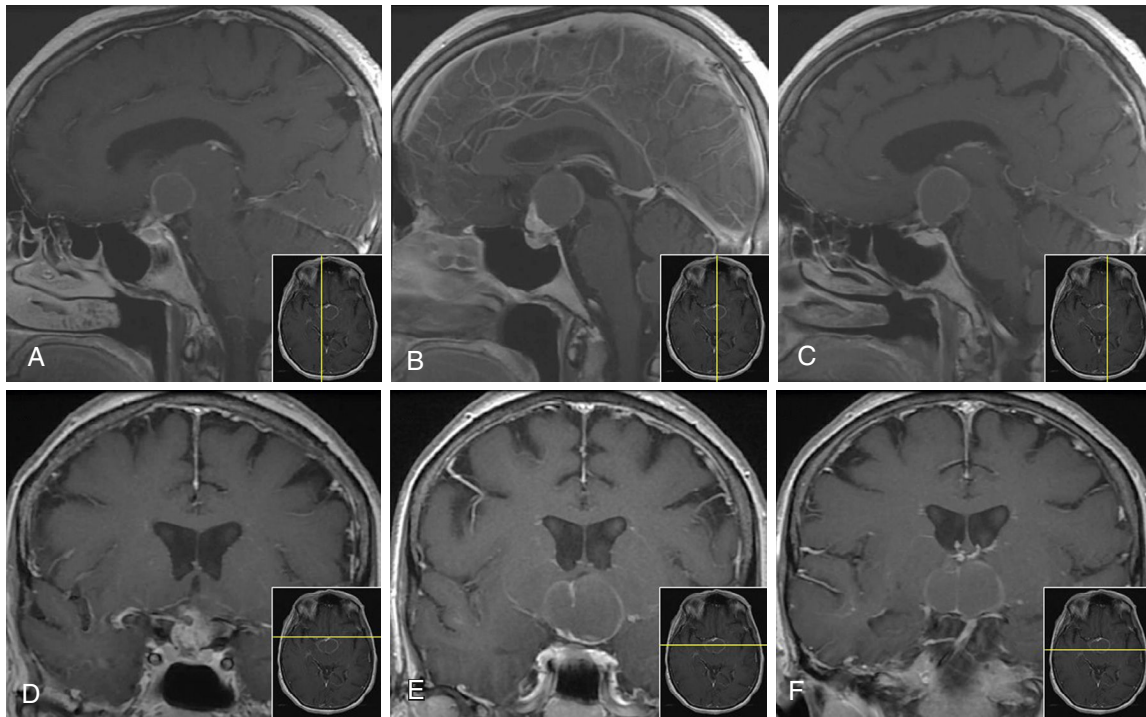


Figure 170.19. A 61-year-old female patient presenting with 2 weeks of lethargy and confusion underwent a craniopharyngioma approach for resection of a craniopharyngioma (A–F).

suprasellar region is approached from a frontolateral direction with identification of the ipsilateral and contralateral olfactory nerves, optic apparatus, and vasculature. An opening is made into the lamina terminalis if the tumor has not already passed through its thin membrane, providing access to the third ventricle. Frequently, this corridor will have been enlarged by tumor invasion, allowing for third ventricular tumor components to be removed more readily.

SURGERY OF THE FOURTH VENTRICLE

Principles

Tumors of the fourth ventricle (Fig. 170.20) can occur in patients of any age, making up 60% to 70% of solid intracranial tumors in pediatric patients.¹⁰⁷ In 1938, Craig et al. noted that early symptoms of fourth ventricular tumors are referable to either or both of two primary causes (mass effect or obstructive hydrocephalus and that early detection and surgical resection led to more favorable outcomes.¹⁰⁸ During this time, Dandy had been performing surgery that he described as “splitting of the cerebellar vermis” to remove fourth ventricular tumors.^{109,110} To this day, the transvermian technique is used to approach the fourth ventricle along with the potentially less destructive telovelar approach described by Matsushima in 1982.¹¹¹

Surgery of the fourth ventricle is challenging with significant risks due to proximity to the facial colliculi, deep cerebellar nuclei, cerebellar vermis, branches of the PICA, cerebellar peduncles, and a number of other neurovascular injuries that could lead to cranial nerve deficits (e.g., oropharyngeal apraxia, weakened cough, diplopia, gastroparesis), gait, or focal motor disturbances. Occlusion of the PICA distal to the medullary branches can result in a syndrome similar to labyrinthitis, causing dizziness, nausea, vomiting, appendicular ataxia, and nystagmus. Manipulation of the floor of the fourth ventricle during surgery may result in hypotension, apnea, or even cardiac arrest. Although advocated by Dandy as a safe approach, incision of the vermis can produce

vermal split syndrome, which presents with oropharyngeal apraxia, cerebellar mutism, disturbed gait, and neuropsychiatric changes.^{112–114} The risk for developing a major neurologic deficit is further compounded in cases of parenchymal tumor invasion, approaching 50% in some series.¹¹⁵

Telovelar Approach

The telovelar approach (Video 170.8) takes advantage of the natural planes found between the cerebellar hemispheres as well as the cerebellum and brainstem. It was developed in response to the transvermian approach, which requires a midline transection or removal of the uvula, pyramid, and tuber. Numerous case reports have described akinetic mutism or abulia following extensive vermian injury, especially in the pediatric population.¹¹⁶ Aside from abating the risks associated with cerebellar vermis split syndrome, the telovelar approach provides greater access to the entire fourth ventricle than the transvermian approach, except for at the superior aspect of the roof, which suffers from a somewhat limited inferior-to-superior angle of exposure.¹¹⁷ However, this can be readily overcome by removing the posterior arch of C1.

Preoperatively, the MR images are inspected for vascular anomalies, especially regarding the vertebral arteries and PICA. An ectatic V3 segment of a vertebral artery may run near the midline, risking injury during soft tissue dissection. In addition, an extradural PICA may occur in up to 20% of patients and can be damaged if not anticipated during operative exposure.¹¹⁸ MRI can also help locate the height of the cerebellar tonsils in relation to the foramen magnum. This helps the surgeon prepare for the degree of hemispheric split that may be required, as well as whether the arch of C1 will require removal to access the rostral aspect of the ventricle. If there is hydrocephalus present, an external ventricular drain may be placed preoperatively to prevent downward herniation upon CSF drainage following dural opening. In our practice, we routinely employ neuromonitoring for this approach. SSEPs, MEPs, BAERs, facial nerve monitoring, and lower cranial nerve monitoring are performed. Some

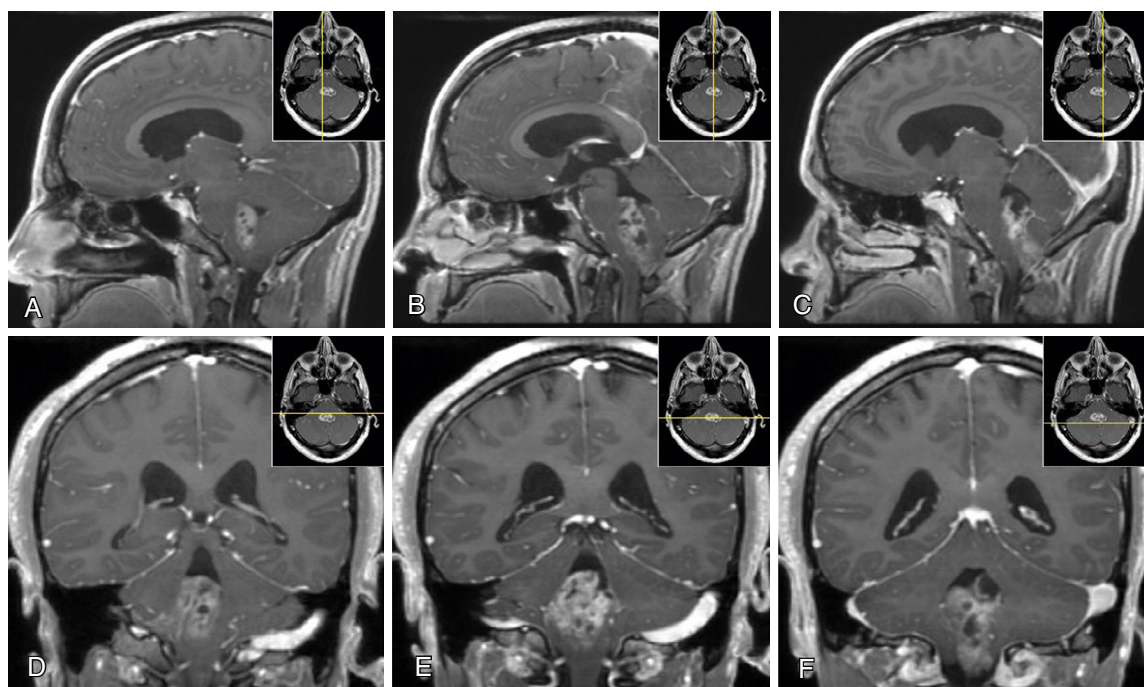


Figure 170.20. A 30-year-old male patient with a 3-month history of dizziness, headache, and nausea underwent complete resection of a grade II ependymoma (A–F) using a telovelar approach.

providers also advocate placement of defibrillator pads because of the risk for asystole or arrhythmia following manipulation of the fourth ventricular floor. The patient's vital signs should be readily visible to the surgeon, and the anesthesiologist should be notified to be completely aware and engaged during resection of the tumor off of the fourth ventricular floor.

For this approach, we prefer to place the patient in the Concorde position to allow for clear orientation of the cerebellar midline and operator comfort. A linear or S-shaped skin incision is performed down to approximately the level of the C2 spinous process, depending on whether the arch of C1 must be removed and on the amount of soft tissue in the patient's posterior neck. Next, the posterior neck musculature is reflected laterally as the dissection is carried down the avascular median nuchal line. Eventually, the superior aspect of the muscles immediately below the superior nuchal line must be divided to allow for adequate lateral retraction. We prefer to spare the attachment of the superior nuchal line, where the muscle meets the occiput, to allow for easier and tighter closure of the posterior musculature. This forms a T-shaped incision within the posterior neck muscles when completed. The muscles are then dissected from the skull and retracted laterally to allow for a suboccipital craniotomy. If the C1 arch does not require removal, we do our best to spare its attachments to the rectus capitis major and minor, provided it does not obscure the surgical view or working angles. We generally choose to perform the midline craniotomy up to the level of the transverse sinus, however exposure of the sinus is not necessary unless a combined supracerebellar approach is required to access the rostral aspect of the tumor because of tumor extension to the cerebellomesencephalic fissure or pineal region. The craniotomy extends down to the level of the foramen magnum, and wide bone removal is performed up to the occipital condyles. However, if tumor extends laterally into the cerebellopontine or premedullary cisterns, the craniotomy can be widened at this point to allow for a retromastoid or far-lateral approach to the anterolateral brainstem. The dura is opened in a standard Y-shaped fashion, extending down to slightly below the level of the foramen magnum. The preoperative MR images should be

inspected carefully to ensure that the PICA is not injured upon dural opening. Sutures are used to reflect the dural folds away from the operative field by suturing them to the surrounding soft tissue. The critical external landmarks for the telovelar approach are now identified. The operator should now have a view of the uvula, pyramid, and tuber of the vermis as well as the bilateral tonsils, retrotonsillar fissures, and the p4 and p5 segments of the PICA. Arachnoid trabeculae running from the cerebellar tonsils to the dura are sharply dissected. This maneuver is repeated for adhesions between the vermis and cerebellar tonsils. Gentle lateral tension on the tonsils reveals any additional points of attachment. Finally, any arachnoid bands that are restricting the lateral retraction of the PICA are removed. Gentle superior and lateral retraction of the cerebellar tonsils allows for exposure of the cerebellomedullary fissure, which contains the uvulotonsillar and tonsillomedullary spaces. With the inferior medullary velum and tela choroidea now under direct visualization, the tela choroidea is opened microsurgically to expose the fourth ventricle, although large tumors will frequently have already eroded through the thin layers of the tela choroidea and inferior medullary velum, making this step unnecessary. Small branches of the tonsillomedullary and telovelatonsillar segment of the PICA run within and beneath the tela choroidea, supplying the choroid plexus, brainstem, and cerebrum. Branches feeding the choroid plexus or tumor can be ligated, while branches to the medulla should be spared. If further exposure of the rostral fourth ventricle is required following the opening of the tela choroidea, an incision can be made that continues upward into the inferior medullary velum. To allow for a wider opening into the lateral recesses, the tela choroidea can be removed up to its point of attachment along the lower posterior wall of the lateral recess.¹¹¹ With adequate tumor exposure achieved, debulking can then commence. It is imperative that natural tumor planes be followed to their interface with the fourth ventricular floor to avoid undue manipulation or damage to normal brainstem structures. If the tumor is invasive into the brainstem without a clear and easily dissectible plane, residual should be left for adjuvant treatment.

INTRAVENTRICULAR TUMOR PATHOLOGY

The aim of this section is to provide a focused overview of pathology that is either commonly found or unique to the cerebral ventricles. It is not intended to provide an exhaustive review of all pathology that can be found within the ventricular system. Knowledge of intraventricular tumor pathology and its classic radiologic appearance is important for guiding next steps in decision making.

Ependymomas

First described by Cushing and Bailey in 1926,¹¹⁹ ependymomas are slow-growing tumors arising from radial glial cells that account for 2% to 9% of intracranial tumors.¹²⁰ There is an increased frequency noted in the pediatric population, peaking to 33% within the first 3 years of life, making them the third most common primary pediatric brain tumor. There is a second smaller peak during the third decade of life.¹⁵ There is no gender bias based on disease prevalence when examined as an entire group, however, various subtypes do vary in their gender distribution.¹²¹ These tumors can arise anywhere within the ventricular system with a predilection for the fourth ventricle (60%), especially in pediatric patients. They can also be found within the spinal canal (10%), typically in adult patients. Interestingly, tumor location correlates with molecularly distinct tumor subtypes.^{121,122} The posterior fossa molecular subtypes include PF-EPN-A and PF-EPN-B (previously posterior Fossa groups A and B), whereas the supratentorial subtypes include the RELA fusion and YAP1 fusion subtypes. PF-EPN-A and RELA fusion tumors typically are found in infants and young children, whereas their PF-EPN-B counterpart is often found in adolescents and adults. Of note, the RELA and PF-EPN-A groups are associated with a poor prognosis. Ependymomas are WHO grade II or III tumors, although anaplastic variants with transformation have been reported in the literature.^{121,123,124} Regardless, no association between grade and biologic behavior has been definitely established.¹²⁵ Patients typically present with hydrocephalus and signs of ICP that correspond to the patient's age. Ataxia, visual disturbances, extraocular eye movement restriction, weakness, and lower cranial nerve deficits are characteristic. Supratentorial tumors may demonstrate focal neurologic deficit or epilepsy.

Grossly, ependymomas are well-defined, lobulated, gray or tan-colored, soft and spongy, frond-like tumors that attach to the ventricular wall by a pedicle, which often contains the tumor's vascular supply. There may be focal areas of calcification, hemorrhage, cysts, or necrosis. These are responsible for the heterogeneous appearance on imaging. Fourth ventricular ependymomas commonly flow through the foramina of Luschka and Magendie to encase the lower cranial nerves and vessels, earning them the name *plastic ependymomas*. As it is not possible to accurately estimate tumor histology or molecular biology based on imaging alone, the need for histopathologic diagnosis and examination is a priority. Classic ependymomas appear similar to well-delineated gliomas, with monomorphic cells of variable density and round to oval nuclei with speckled nuclear chromatin. Cell density can vary considerably, especially in the posterior fossa variants. Perivascular rosettes and true ependymal rosettes are key features on histology, although ependymal rosettes are found with much lower frequency. Other histologic features include regions of myxoid degeneration, intratumoral hemorrhage, dystrophic calcification, and, rarely, metaplastic cartilage or bone. Ultrastructurally, ependymomas retrain their ependymal cilia, blepharoblasts, and microvilli. Ependymomas stain positive for glial fibrillary acid protein (GFAP), epithelial membrane antigen (EMA), S100, and vimentin. OLIG2 expression is characteristically low compared with other gliomas. Molecular alterations are common in ependymomas. Important variations

include the homozygous deletion of *CDKN2A* in supratentorial tumors as well as chromosome 1q gain in posterior fossa tumors, both indicating a poor outcome.

On CT imaging, ependymomas are isodense to parenchyma and enhance heterogeneously with areas of calcification or recent hemorrhage appearing hyperdense. On MRI, ependymomas appear as well-circumscribed masses that are hypointense to isointense to brain parenchyma on T1-weighted imaging with heterogeneous enhancement. On T2-weighted sequences, the tumor appears hyperintense to white matter. Occasionally, restricted diffusion may be seen on diffusion-weighted imaging. Supratentorial ependymomas typically demonstrate cystic components. Gross brain invasion or edema are exceedingly rare. If an ependymoma is suspected on imaging or confirmed by histology, MRI of the entire neuroaxis should be obtained because of the known risk for seeding within the CSF pathways, and it is a key component for staging, prognostication, and treatment. There is a well-established association between NF2 and spinal intramedullary tumors.

Key factors for prognosis include patient age, tumor location, molecular subtype, and extent of surgical resection. Patients younger than 12 months of age demonstrate a poor 5-year survival rate of around 40%.¹²⁶ This improves with age to about 55% in patients 1 to 4 years of age and greater than 75% in patients older than 10 years of age. Supratentorial and spinal ependymomas have significantly more favorable outcomes than posterior fossa neoplasms, particularly in children. Metastatic disease is indicative of poor prognosis. As mentioned previously, the RELA fusion and PF-EPN-A groups are notorious for having a poor prognosis. Surgical resection is critical for the treatment of ependymomas, with extent of resection representing a risk factor independent from pathohistologic grade.^{127,128} In pediatric and adolescent patients, 10-year overall survival of up to 80% following gross total resection and radiotherapy has been observed.^{15,129,130} In cases of recurrence, the prognosis is very poor.¹³¹ At this time, chemotherapy is not an established adjuvant therapy for children or adults after index surgery for their ependymoma, although it may be used in cases of tumor recurrence. However, infants younger than 1 year of age are typically bridged to radiation treatment using chemotherapy. This delays the deleterious effects of radiation on the developing nervous system.^{132,133} At around the 1-year age mark, these patients typically undergo surgery for any residual disease followed by radiotherapy if necessary. In older patients, radiotherapy is considered for cases of subtotal resection, high-grade tumors, or tumor dissemination throughout the neuraxis, for which it has been shown to improve survival.¹²⁹

Subependymomas

Subependymomas were first described by Scheinker in 1945, typically being diagnosed incidentally or on autopsy.¹³⁴ They are rare and occur mostly in adults in their fifth and sixth decade of life, with a 2 : 1 male gender predilection. Subependymomas account for just 0.2% to 0.7% of intracerebral tumors, with the majority occurring within the fourth (60%) and lateral ventricles (30%).^{15,121,135,136} Less common sites include the third ventricle, septum pellucidum, brain parenchyma, or cervicothoracic spine. Generally a benign and slow-growing ventricular wall tumor arising from subependymal glial or astrocytic cells, subependymomas are rarely invasive, lending to their WHO I classification.¹²¹ Because of their relatively benign nature, they are most often discovered incidentally or in patients with obstructive hydrocephalus, although intratumoral hemorrhage and seizures have been reported.¹³⁵ Brain stem compression from large fourth ventricular tumors has been described in both adults and children. With the growing availability of imaging studies, these tumors are being diagnosed with increased frequency.^{15,137}

On gross appearance, subependymomas appear as sharply demarcated nodules attached to the ependymal lining. Large tumors may have cystic or calcified components.¹³⁸ On histopathology, these tumors display low cellularity and lack high-grade features such as mitotic figures, vascular endothelial proliferation, or necrosis. Loose perivascular ependymal pseudorosettes are occasionally seen, as well as a dense fibrillary matrix and small cysts. Cell nuclei generally appear isomorphic but are variable in cases of multicystic tumors. At the ultrastructural level, subependymoma cells may show typical ependymal characteristics such as cilia, microvilli, and intermediate filaments. Sometimes, a higher-grade ependymoma component is present, with the tumor behaving more similar to the high-grade specimen.^{15,121} Cells may express GFAP, but unlike ependymomas do not stain for EMA. Potential targets include TOP2B, MDM2, nucleolin, HIF1- α , and phosphorylated STAT3, all frequently expressed in subependymomas.

On CT imaging, subependymomas appear hypodense to isodense compared with parenchyma, and they do not enhance. Occasionally cystic or calcified areas may be noted on imaging.¹³⁸ T1-weighted imaging typically reveals tumor nodules that appear hypointense or isointense relative to white matter with minimal enhancement. On T2-weighted imaging, subependymomas often appear hyperintense relative to neighboring structures but do not usually demonstrate cerebral edema.

As these tumors are benign in nature, routine follow-up with surveillance MRI is generally favored in the absence of clinical signs or symptoms. However, in the case of rapid growth or the development of new symptoms, surgical resection should be considered. Given the benign nature of the tumor, complete resection of the tumor defines surgical cure. In general, adjuvant therapy is not required but may be considered in cases of significant tumor residual. This occurs most often for tumors adherent to the fourth ventricular floor. To date, no recurrences after gross total resection have been reported in the literature.¹²¹

Surgical removal of subependymomas is technically straightforward. These tumors are particularly vascular or invasive. We favor gross total resection if safely possible, especially in younger patients, given the chance for a curative procedure. Total removal may spare patients the need for radiotherapy in the event of tumor progression.

Central Neurocytomas

Not established as a unique tumor entity until 1982 by Hassoun et al. following the advent of electron microscopy, these tumors are exceedingly rare, representing only 0.1% to 0.5% of intracranial tumors.^{15,121,139} Most commonly found in young adult patients between 20 and 40 years of age, there is no gender bias for these neuroepithelial tumors. Initially classified as WHO grade I tumors, this was changed to grade II in 1993 as a number of cases have exhibited more aggressive behavior.¹⁵ These tumors demonstrate a predilection for the lateral ventricles, specifically the frontal horn of the left ventricle, and often maintain an attachment to the septum pellucidum. Cases of third or even fourth ventricular location have been reported as well, but with very low frequency. Third ventricular location often arises as a result of extension from the lateral ventricles through the foramen of Monro. Although the cell of origin is unknown, a neuroglial precursor with dual differentiation potentiality is often implicated because of the mixed glial and neuronal components. Despite their generally slow growth pattern and thus insidious onset of symptoms, there have been reports of sudden death caused by acute ventricular obstruction or hemorrhage.¹⁴⁰ Presentation classically is that of increased ICP.

Macroscopically, central neurocytomas appear gray and friable, and they demonstrate areas of calcification and hemorrhage.¹²¹ Microscopically, cells are typically uniform and round with finely

speckled chromatin. There are often areas of variable architecture similar in appearance to oligodendrogliomas, pineocytomas, or neuroendocrine tumors. These histologic similarities to other neoplasms have led to a late establishment of this tumor's own identity. For this reason, the use of immunohistochemical markers and genetic testing are key to the definitive diagnosis. Given their neuronal origin, central neurocytomas stain positive for neuronal markers including synaptophysin, NeuN, neuron-specific enolase, MAP2, and class III β tubulin. Their genetic profile often contains numerous DNA copy number alterations including MYCN gain and WNT signaling pathway overexpression. Unlike oligodendrogliomas, neurocytomas lack *IDH* mutations or 1p19q co-deletions. Currently MIB-1 is the most accurate means of determining prognosis, likelihood of tumor relapse, and tumor grade.

On CT imaging, central neurocytomas are usually slightly hyperdense or isodense compared with brain parenchyma, with over one-half exhibiting punctate calcifications.^{15,140} Moderate enhancement can be seen with iodinated contrast on CT. There are often cystic regions found on large tumors. On MRI, neurocytomas appear isointense to gray matter on T1-weighted sequences, also demonstrating mild to moderate enhancement. Tumors are typically hyperintense to brain on T2-weighted and FLAIR imaging, with a "soap bubble" appearance caused by their cystic components. There is typically a well-defined margin around the tumor. Solid components may demonstrate diffusion restriction. Cases of CSF dissemination have been reported, however this remains the exception.¹⁵ Unfortunately, radiographic differentiation from other tumors such as oligodendrogliomas is exceedingly difficult.¹⁴¹⁻¹⁴³

Complete surgical excision of neurocytomas is frequently curative and is the "gold standard" treatment for central neurocytomas.¹⁴² The 5-year survival was greater than 88% in multiple series in which gross total resection was achieved.^{141,144-147} In the event of subtotal resection, around one-half of central neurocytomas recur within 5 years. Adjuvant radiotherapy has been shown to increase progression free and overall survival in cases of incomplete resection, however some practitioners prefer to observe for recurrence with a plan for a second operation before adjuvant therapy.^{142,147-149} The use of chemotherapeutic treatments is still under debate, although it has thus far demonstrated limited efficacy.^{141,150,151}

Pilocytic Astrocytomas

Pilocytic astrocytomas account for 5.4% of all gliomas, found most commonly in patients younger than 20 years of age. The most common pediatric glioma, it accounts for one-third of gliomas in children 0 to 14 years of age and 18% of all primary childhood brain tumors. There is no gender bias for patients affected by this WHO grade I tumor. Symptoms of increased ICP are common among infants and young children. Lesions affecting the optic pathway or hypothalamus can cause blindness or endocrinologic disturbances, most often obesity or diabetes insipidus. Cerebellar tumors may present with ataxia, clumsiness, headache, nausea, or vomiting. Hemiparesis may be seen in thalamic tumors compressing or infiltrating the internal capsule. Cranial nerve and long-tract deficits can be observed if the brainstem is involved. Anaplastic transformation as well as leptomeningeal seeding are rare but have been reported.

On gross examination, pilocytic astrocytomas appear soft, gray, and typically discrete from surrounding tissue. Intratumoral or paratumoral cyst formation is frequent, often accompanied by a mural nodule. Microscopically these tumors demonstrate various tissue patterns, sometimes containing several tissue types in a single lesion. The cerebellar variant most commonly contains the classic biphasic pattern, with variable proportions of compact bipolar cells with Rosenthal fibers and loose-textured microcystic

multipolar cells. Some common findings include Rosenthal fibers, eosinophilic granular bodies, dense vasculature, and regressive features. Staging is determined by the various tissue types. Mutations in the *BRAF* and *NF1* genes are common in pilocytic astrocytomas. As with other low-grade pediatric tumors, these often lack *IDH* and *TP53* mutations. These astrocytic tumors strongly express GFAP, S100, and OLIG2.

On imaging, lesions are typically well demarcated, midline, and cystic with smooth margins. An avidly enhancing mural nodule on contrasted CT imaging is frequently found. On MRI, the tumors appears similar, with a cystic component and an enhancing mural nodule. Variations can include a mass with a nonenhancing central area or a solid mass without a cystic portion. NF1-associated tumors often originate along the optic apparatus and are typically not cystic. Calcifications are rare, occurring in only one-fifth of tumors.

As these tumors are often exceedingly slow growing, complete resection is core to the patient's treatment plan. The 10-year survival rate following treatment has been cited as greater than 95%. Given complete surgical removal, neither radiotherapy nor chemotherapy is required. Radiotherapy is avoided in the developing infant to prevent cognitive and other developmental delays.

Choroid Plexus Tumors

Choroid plexus papillomas are neuroectodermal tumors accounting for 0.5% to 0.6% of intracranial lesions in adults and 10% to 20% of tumors in infants younger than 1 year of age. They arise from the cuboidal epithelium lining the choroid plexus papillae, occurring in similar frequencies in both men and women. These are among the most common tumors in children younger than 2 years of age, with 85% of choroid plexus papillomas occurring in children younger than 5 years of age.¹²¹ Patients present with signs of hydrocephalus in over 80% of cases¹⁵² resulting from either CSF overproduction or CSF pathway blockage. In children, this may present as developmental delay, behavioral abnormalities, or epilepsy. Most of these tumors are located in the ventricular atrium of the lateral ventricles in pediatric patients. In adults, a fourth ventricular location is most common. Presence of this tumor in the third ventricle, cerebellopontine angle, or pineal region has been documented but is exceedingly rare. Malignant transformation in recurrent or incompletely resected tumors has also been described.¹⁵³ Known associations include Aicardi syndrome, Down syndrome, von Hippel-Lindau disease, and neurofibromatosis type 2.^{154–156} The simian virus 40 (SV40) has been associated with choroid plexus carcinomas and is present in up to 50% of carcinomatous tumors.¹⁵⁷

Macroscopically, choroid plexus tumors appear well circumscribed and globular, and they are often described as “cauliflower-like” in appearance. They frequently have a vascular stalk and may adhere to the ventricular wall. They appear pink to red-brown because of their extensive vascularity. Carcinomas may exhibit areas of necrosis and are more likely to invade the surrounding parenchyma. Under microscopic examination, choroid plexus papillomas demonstrate papillary structure with a delicate fibrovascular core lined by columnar or cuboidal epithelial cells with vesicular basal nuclei similar to that of the normal choroid plexus.¹⁵ They stain positive for cytokeratins, vimentin, KIR7.1, and transthyretin. S100 staining is variable, however its presence is a positive prognostic factor. Of note, characteristic additions and deletions separate from that of choroid plexus carcinomas suggest the possibility of a separate genetic basis for each of these tumors. The most common underlying genetic mechanism identified in choroid plexus carcinomas is TP53 dysfunction. Compared with choroid plexus papillomas, atypical choroid plexus papillomas show increased mitoses, nuclear pleomorphism, and occasionally necrosis. Choroid plexus

carcinomas have an even higher mitotic rate, increased cellularity, and blurred papillary structure. Also, they frequently exhibit necrosis and brain parenchymal invasion. These WHO grade III tumors often do not stain for S100, transthyretin, or EMA.

CT imaging most often demonstrates a well-defined and lobulated mass that is isodense to adjacent brain parenchyma. Just as the choroid plexus enhances avidly with delivery of contrasted dye, choroid plexus tumors demonstrate marked enhancement, appearing almost “frond-like.” Occasionally, in about one-fourth of cases, calcifications may also be noted. On T1 MRI sequences, tumors appear isointense to white matter. T2-weighted imaging often reveals a slightly hyperintense tumor, with surrounding edema in the case of parenchymal invasion. Tumors enhance vividly because of their vascular nature, with small areas of flow voids. Although it can be difficult to differentiate a choroid plexus carcinoma from lower-grade tumors, the most apparent distinction is the presence of parenchymal invasion. If a choroid plexus carcinoma is suspected, imaging of the entire neuraxis should be conducted because of the risk for seeding.

For choroid plexus papillomas, complete surgical resection is the treatment of choice to provide local control. Overall survival for this patient group is excellent, with some series demonstrating a greater than 95% 10-year overall survival.^{13,158–160} Unfortunately, survival in the case of choroid plexus carcinomas is significantly worse, typically around 40% at 5 years.^{13,157,158} The use of adjuvant radiotherapy and chemotherapy (ifosfamide, carboplatin, and etoposide) has been shown to benefit patients with more aggressive carcinomatous variants.^{161,162} Neoadjuvant chemotherapy has been shown to benefit ease of surgery and can be used in cases of recurrence. These adjuvant treatments are often not necessary in run-of-the-mill choroid plexus papillomas and are avoided in pediatric patients if safe surgical resection can be performed because of their developmental side effects.

When operating on choroid plexus papillomas, the operator should be prepared to encounter a highly vascular tumor with the potential for developing intraoperative or postoperative hemorrhage. It is advised to use imaging to identify arterial feeders, usually from choroidal vessels, to allow for early devascularization. After this is achieved, the tumor can be removed using a combination of bipolar coagulation, suction, and ultrasonic aspiration. The neighboring choroid plexus is coagulated and removed to reduce the risk for recurrence.

Meningiomas

The term *meningioma* was first introduced by Harvey Cushing in 1922.^{163,164} Since then, meningiomas have been found to be the most common primary brain tumor, accounting for approximately one-third of all primary brain and CNS tumors.^{121,165} Intraventricular meningiomas are significantly less common, making up less than 2% of all meningiomas, yet they are among the most common types of intraventricular tumors.¹⁴⁰ Theorized to arise from meningeal inclusion bodies within the tela choroidea or choroid plexus stroma, these tumors exhibit a 2 : 1 female predominance and are generally found in adults in the fourth to sixth decades of life.^{166,167} The majority of ventricular meningiomas are found in the lateral ventricle atrium, with a left ventricular predominance.^{168,169} Occurrences in the third and fourth ventricles are significantly rarer still. The WHO grading of meningiomas ranges from I to III, with an exceedingly wide variety of histologic subtypes. Intraventricular meningiomas are most commonly fibrous meningiomas (WHO grade I),^{121,166} although reports of more aggressive variants have been reported in the literature.^{170–172}

Imaging findings of intraventricular meningiomas are similar to those of meningiomas found at other locations. CT imaging reveals a mass that is either slightly hyperdense or isodense to surrounding brain tissue and usually enhances vividly with

contrast administration. Intraventricular meningiomas are more frequently calcified than their extraventricular counterparts, occurring in about 50% of lesions. On MRI, meningiomas are typically hypointense or isointense on T1-weighted sequences compared with gray matter, and they enhance avidly. On T2-weighted imaging, meningiomas often appear hyperintense or isointense to surrounding gray matter. They generally are well circumscribed and may demonstrate a spoke-wheel appearance. Edema from intraventricular tumors is rare, and its presence anecdotally suggests a more aggressive subtype or brain invasion. However, this has not been confirmed in the literature. Intraventricular meningiomas are relatively vascular tumors that are fed by choroidal vessels. Early devascularization is key for minimizing intraoperative blood loss during surgery. Given the nature of the choroidal feeders, we do not utilize embolization for removal of these tumors. Drainage to deep veins, often the internal cerebral veins, is common. The use of phased dynamic CTA allows for more precise localization of key feeding and draining vessels if deemed necessary.¹⁶

Macroscopically, intraventricular tumors appear similar to their extraventricular counterparts. They typically are tan or gray in appearance, round and lobulated, and well circumscribed. They often follow the contours of the ventricle that they inhabit. Microscopic findings such as loculated architecture containing meningothelial whorls, psammoma bodies, syncytial cells with indistinct membranes, eosinophilic cytoplasm, uniformly round nuclei with pseudo-inclusions, and xanthomatous degeneration may point toward the diagnosis of meningioma. Tumors typically stain for vimentin, EMA, and ProgR. Focal staining with S100 may be present. Proliferative markers such as Ki-67, anti-phosphohistone-H3 (PHH3), and apoptotic index have been shown to correlate with biologic behaviors.^{173–175} Recently, significant work has been done to identify genetic mutations associated with meningiomas. Findings have been tied to location and subtype and have allowed for the identification of novel therapeutic opportunities.^{176–180} The *NF2* mutation, already the most frequently found mutation in meningiomas, is found with even greater frequency in this population.

Total resection is a curative treatment for symptomatic meningiomas. Asymptomatic individuals may be monitored with serial imaging and followed clinically for the development of neuropsychiatric changes. If hydrocephalus or cognitive deficits arise, surgical options should be considered. The origin of intraventricular tumors from meningeal inclusion bodies within intraventricular structures, rather than arachnoid cap cells, invalidates the classic Simpson Grading schema when determining the rate of recurrence.¹⁸¹ Given their origin as inclusion bodies within the choroid or tela choroidea, we are careful to resect and coagulate these structures as much as is safely possible to reduce the possibility of recurrence. The use of an endoscope to confirm gross total resection is key for minimizing residual tumor. Patients generally have a favorable prognosis following tumor removal.^{182–188} Residual tumor and recurrence may be treated with radiotherapy to improve local tumor control.^{189,190}

Our approach to the resection of intraventricular meningiomas focuses on balancing early devascularization and optimal working angles with minimizing brain retraction and maximizing the use of natural corridors. In the ideal setting, early devascularization of the choroidal feeder can take place, followed by debulking of the tumor mass and finally removal of the outer tumor and its capsule. We prefer to operate along the long axis of the tumor whenever possible to avoid the need to perform significant brain retraction when trying to reach contralateral recesses that are hidden from view. If the tumor is suspected to be invasive, vascular, heavily calcified, or appears particularly adherent, we avoid any approach that requires a long reach with limited working angles that may compromise our ability to safely perform surgery.

Epidermoid Cysts

Intracranial epidermoid cysts make up about 1% of all intracranial tumors, with a fraction (~25%) involving either the fourth ventricle or suprasellar region.¹²¹ Epidermoid cysts originating in the lateral or third ventricles are exceedingly rare. Epidermoid cysts stem from inclusion of ectodermal elements during neural tube closure or following trauma, and they typically present with symptoms of mass effect or hydrocephalus when originating from the ventricles. Cysts typically are diagnosed in patients between their second and fourth decades of life, with a slight male preponderance.^{191,192} Malignant degeneration is exceedingly rare.¹⁹³

Radiographically epidermoid cysts appear isodense to CSF on CT imaging because of their combination of cellular debris and high cholesterol content. They do not enhance with contrast administration. MRI yields similar results, appearing isointense on T1- and T2-weighted sequencing. This radiographic profile is identical to that of an arachnoid cyst. The two entities are best differentiated using diffusion-weighted imaging/apparent diffusion coefficient (DWI/ADC) sequences. Epidermoid cysts exhibit restricted diffusion because of their hypercellular nature, while arachnoid cysts do not.

On gross inspection, epidermoid cysts are encapsulated, round, white masses with thin, white walls and soft, pearly-white keratin sheets. There is no mistaking the appearance of their creamy and waxy material. Microscopically, the cyst contains anucleate squamous epithelium without skin adnexae or hair, which would be characteristic of a dermoid cyst or teratoma. Cells stain positive for keratin and EMA. There is an uncommon association with anorectal malformation, sacral agenesis, and the presence of a presacral mass (frequently epidermoid tumor), which is known as the *Currariano triad* and is often associated with *MNX1* mutations.¹⁹⁴

Complete surgical removal of epidermoid cysts yields excellent disease control. Removal of the cyst capsule is generally advised to reduce the risk for cyst recurrence and postoperative aseptic meningitis.^{195,196} Radiotherapy has also been shown to decrease the risk for progression and recurrence, although this is not standard care.^{197,198} In cases of malignant degeneration, chemotherapy has been utilized with mixed results.¹⁹³ The goal for the vast majority of cases, which are benign in nature, should be complete surgical extirpation to avoid the need for further adjuvant therapy.

Dermoid Cysts

Dermoid cysts are a rare occurrence, making up less than 0.5% of primary intracranial tumors.¹⁹⁹ Ectodermal in origin, they result from ectopic tissue located within the meninges, ventricles, or brain parenchyma. They are found more frequently in females and in patients during the first three decades of life.²⁰⁰ Patients typically present with intermittent severe headaches caused by bouts of aseptic chemical meningitis resulting from spontaneous or traumatic rupture. Presenting symptoms occasionally occur from hydrocephalus or local mass effect (e.g., tectal or optic nerve compression).^{201,202} Although generally aseptic in nature, isolated reports of infection following rupture have been published.^{203–206} Rarely, dermoid cysts have been implicated in seizures, vasospasm, or death.^{207,208} Dermoid cysts are typically midline and infrequently present with a fistulous tract that extends extracranially to the dermis.

On CT imaging, dermoid cysts demonstrate hypoattenuation because of their high fat content. Frequently these areas of low attenuation appear lobulated, and enhancement is uncommon. Ruptured dermoid cysts characteristically demonstrate fat-attenuating material scattered throughout the subarachnoid space. On T1-weighted sequences, intracystic fat appears hyperintense,

while the T2-weighted appearance is variable. Enhancement is minimal, when present.²⁰⁹

Dermoid cysts have relatively thicker walls than epidermoid cysts and contain a greasy pilosebaceous content. Variable tissue types including hair and fat globules can be noted during surgery. Microscopically these tumors can demonstrate a fibrous wall and keratinizing squamous epithelium that includes skin adnexa, squamous cells, hair, and sebum. In the case of prior rupture, a foreign body giant cell reaction may present. Like epidermoid cysts, these tumors stain positive for keratin and EMA. Hair shafts can be readily seen using polarized light.

Surgical removal of these benign tumors is the treatment of choice, especially in the case of recurrent rupture, seizures, severe headache, hydrocephalus, or symptomatic local compression. Similar to epidermoid cysts, removal of the tumor wall is advised to reduce the risk for recurrence and postoperative meningeal irritation.¹⁹⁶ It is rare for malignant transformation to occur.¹⁹²

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